Quantum Mechanics

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Preface

Welcome to our Quantum Mechanics book!

What is This?

This book contains a comprehensive catalog of the fundamentals of quantum mechanics (QM), to the extent that the authors have encountered the topics and the material. With this book, our objective is to provide an intuitive approach to learning quantum mechanics, and share with you the tips and tricks we learned while we were studying the field. Whether you're just getting started with QM or are already familiar with some of the content covered, we hope there is something here for you to learn!

Fundamental Structure

We have divided this book into three major parts, in an attempt to effectively compartmentalize the vast amount of material we aim to cover. The first part covers fundamental theory; specifically, it details the basic building blocks of quantum mechanics and helps us transition from the classical frame to the quantum one. The second part shifts focus to approximation methods. As we will learn, solving quantum systems exactly is nigh-impossible, which necessitates the need to approximate in order to effectively study physical systems. The final part will explore some more advanced special topics and extend beyond quantum mechanics into the relativistic frame.

How to Use This Book

The purpose of this book is to provide the reader with the foundations for a basic understanding of quantum mechanics. We will attempt to be as rigorous as possible when appropriate, but we stress intuitive understanding above anything else. With that said, this book is not an adequate substitute for any other reading materials or a university course. This, first and foremost, should be a springboard for further discovery. After reading a chapter, the hope is the reader has a general understanding of what was covered, to the extent that they can venture out on their own and supplement their knowledge with additional resources.

Some Notational Logistics

Before we get started with the content, let's go over some basic elements you will see throughout this book.

Definition 0.1 (A Thing)**.** Whenever we come across a new term, we will define it like so.

We will also occasionally highlight specific mathematical notation like so.

Lemma 0.1 (Auxiliary Trick)**.** *If there's a relevant fact that services a larger result, we will encode it as a lemma.*

Theorem 0.1 (Major Result)**.** *When we arrive at a major result, we will encode it as a theorem.*

Proof. We will take care to prove theorems as soon as possible. □

Corollary 0.1 (Consequence)**.** *If a theorem has an important consequence, we'll label it as a corollary.*

Remark. If there's an important side note we wish to highlight, we'll place those here.

Example 0.1 (What Just Happened)**.** Examples will help us illustrate what just happened.

Read these! These contain common mistakes and things to be careful.

We will try to keep such remarks and warnings to a minimum. We will also try our best to keep theorems and important results cross-referenceable; for instance, clicking Theorem 0.1 will take you to the corresponding theorem (or hovering over it will give you a summary). Finally, this book combines elements from various books that we've encountered over our own journey through quantum mechanics, a full list of which can be found in the references page.

One last thing: there is a lot of content in this book, and there are only two of us actively maintaining it. If you find that there are any errors, or you wish to make any suggestions, please reach out via email and let us know! We both want this book to be as approachable as it can be, and are always looking for ways to improve it. You can find our emails in the author list at the top of the page.

Happy reading!

Part I

I: Fundamental Theory

The first part of our book will cover some fundamental theory to establish the general framework of quantum mechanics. We will provide some motivation for why quantum mechanics is needed/how it historically arose, establish a formal mathematical framework which we will use to develop theory, acquire some useful tools that will build the foundation of the field, and examine how to model some basic quantum mechanical systems with said tools.

We have a long journey ahead of us, but don't worry; we will take great care to gently (but thoroughly) introduce the relevant tools for you (the reader) throughout each chapter. Without further delay, on with the show…

1 History of Quantum Mechanics

In order to dive into the world of quantum properly, it pays to spend some time looking at its history. In this chapter, we will give a brief overview of how quantum mechanics came to be. This way, you will appreciate not only the beauty of quantum mechanics, but also understand why it is necessary to understanding the mechanics of our world.

The content in this section is heavily inspired by Prof. Siddiqi's notes for the introductory lecture in Physics 137A (Introductory Quantum Mechanics) at UC Berkeley.

1.1 Blackbody Radiation

To begin our story, we start in the mid 19th century. Around this time, Gustav Kirchhoff (yes, the one from your circuits classes) was studying the physics of black bodies — an idealized object that absorbs *all* electromagnetic radiation, hence the term "blackbody". Even though perfect blackbodies do not exist in the real world, this theory turns out to be a very good model for celestial bodies like planets and stars.

Kirchhoff's crucial insight on blackbodies was that their emissive power R was only determined by two factors: the emitted wavelength λ, and its temperature T. At around the same time, Josef Stefan empirically determined that the total radiation emitted by a blackbody is proportional to T^4, meaning that the total power across all wavelengths, as a function of temperature, may be written as:

$$R(T) = \int_0^\infty R(\lambda, T)\, d\lambda = \sigma T^4 \tag{1.1}$$

The constant $\sigma \approx 5.68 \times 10^{-8}$ is the **Stefan-Boltzmann constant**. While this relation was known, the functional form of $R(\lambda, T)$ eluded physicists at the time. In the early 20th century, Lord Rayleigh and Sir James Jeans used a purely classical approach and arrived at the *Rayleigh-Jeans law*, given as:

$$R(\lambda, T) = \frac{8\pi k_B T}{\lambda^4}$$

Here, $R(\lambda, T)$ denotes the power emitted per unit area. But, there is a problem with this formula! This formula predicts that at smaller and smaller λ, objects should emit an enormous amount of energy, due to the λ^4 scaling in the denominator. Putting this into Equation 1.1

would result in an infinite amount of radiated energy, as the integral diverges at $\lambda \to 0$. This is clearly not physically true (the sun is at finite temperature, and does not radiate infinite energy), and this discrepancy is what is known as the **ultraviolet catastrophe**. This is the first indication that classical physics was incomplete.

1.2 The Quantization of Energy

The resolution, as it turns out, is to introduce the idea of **quantization**. To fully appreciate why this is needed, let's take a small detour. First, consider a square box of length L, and we want to calculate the number of standing modes in the cavity. These waves must satisfy the wave equation:

$$\nabla^2 \Psi(r,t) = \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\psi(r,t)$$

The solutions to this differential equation are sine waves, which in three dimensions are functions of the form:

$$\psi(r,t) = A(t)\sin(k_x x)\sin\left(k_y y\right)\sin(k_z z)$$

Then, since we require that the wave is zero outside the cavity, we enforce $\psi = 0$ outside the cavity. We express this constraint mathematically as requiring that:

$$k_i = \frac{n_i \pi}{L}$$

Now, to solve for $A(t)$, we plug this back into the wave equation, which gives us a solution of the form:

$$A(t) = A_0 \cos(\omega t) + \phi$$

The ϕ factor is simply a phase which we don't care that much about — we can always zero this out for simplicity. What's more important here is ω, which from our earlier condition on k satisfies the equation:

$$\omega^2 = \frac{c^2\pi^2}{L^2}\left(n_x^2 + n_y^2 + n_z^2\right) \tag{1.2}$$

From this equation, you can then see that the number of ways n_x, n_y and n_z can be chosen increases with increasing ω, which we will explore in the next section.

1.2.1 Number of Modes

Now that we have the full form for $\psi(r,t)$, let's actually figure out how many modes there are for a given ω — in other words, for a given wave energy ω, what is the number of allowed modes $N(\omega)$ inside the box? To do this, we first define a *density of states* function $g(\omega)$, defined as:

$$g(\omega) = \frac{\mathrm{d}N(\omega)}{\mathrm{d}\omega}$$

This way, we have

$$N(\omega) = \int_0^\omega g(\omega)\ d\omega$$

For a fixed ω, $g(\omega)$ is implicitly given by Equation 1.2: it is equal to the number of ways to choose n_x, n_y, n_z such that the left and right maintain equality. Since $N(\omega)$ integrates from 0 to the maximum allowed ω, then we are essentially looking for the number of ways to choose the n_i such that the right hand side of Equation 1.2 doesn't exceed the left. Written explicitly as an inequality, we are basically looking for:

$$n_x^2 + n_y^2 + n_z^2 \leq \frac{\omega^2 L^2}{c^2 \pi^2}$$

This equation happens to be relatively easy to solve: the right hand side defines a sphere of radius $\frac{\omega L}{c\pi}$, and since n_i must be positive, we must only consider the octant where each $n_i > 0$, which is $\frac{1}{8}$-th of the volume of the sphere. As such, $N(\omega)$ is given by:

$$N(\omega) = \frac{1}{8}\left(\frac{4}{3}\pi \frac{\omega^3 L^3}{c^3 \pi^3}\right) = \frac{\omega^3 V}{6c^3\pi^2}$$

Here, $V = L^3$ denotes the volume of the box. Converting this to linear frequency using $\omega = 2\pi f$, we get:

$$N(f) = \frac{8\pi^3 f^3 V}{6c^3\pi} \implies g(f) = \frac{\mathrm{d}N(f)}{\mathrm{d}f} = \frac{4\pi f^2 V}{c^3}$$

Since each mode contains two different polarizations (you can loosely think of this as linearly and circularly polarized light), we have to multiply $g(f)$ by 2 to get the full expression:

$$g(f) = \frac{8\pi f^2}{c^3} V$$

So far, this derivation is purely classical, and it is motivated purely by the mathematics. As we will see, this is where the similarities in the classical and quantum treatment end.

1.2.2 Finding the Energy Density

When Rayleigh and Jeans derived their formula, they assumed that all modes exist and have energy $k_B T$, a property given by the equipartition theorem.[1] Using this assumption, we then know that the number of modes between f and $f + df$ is given by

$$g(f) k_B T\ df = \frac{8\pi}{c^3} f^2 V k_B T\ df$$

[1]The equipartition theorem says that the average of any quadratic degree of freedom is $\frac{1}{2}k_B T$. In short, the reason why each mode contributes $k_B T$ is because the Hamiltonian for harmonic oscillators have two quadratic degrees of freedom: one in position and another in momentum. Each contributes $\frac{1}{2}k_B T$, so in total each mode contributes $k_B T$. For a more detailed explanation, consult a statistical mechanics book.

The energy density is given by dividing out the volume V:

$$\rho(f) = \frac{8\pi}{c^3} f^2 k_B T \; df$$

Changing this into an expression involving λ gives:

$$\rho(\lambda) = \frac{8\pi}{c^3} \frac{c^2}{\lambda^2} \frac{c}{\lambda^2} k_B T = \frac{8\pi k_B T}{\lambda^4}$$

which is precisely the energy density given by the Rayleigh-Jeans law, giving us the ultraviolet catastrophe. However, if we instead assume that not *all* modes exist but that energy is **quantized**, we get a different result. This is the assumption that Max Planck made, and it is what allows us to arrive at the Planck radiation formula. In particular, Planck assumed that energy may only be gained and lost in units of hf (called *quanta*) where h is Planck's constant. With this assumption, it is possible to come up with a different energy density equation, now known as the **Planck radiation formula**:

$$\rho(\lambda, T) = \frac{8\pi hc}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}$$

The derivation for this equation is rather complicated, and involves the heavy use of statistical mechanics, so we will skip it here. What's important is that this theory, unlike the Rayleigh-Jeans law, agrees very well with experimental data.

So what's the takeaway from all this? That we *had* to take quantum effects into account in order to come up with the correct relation for $\rho(\lambda, T)$.
Hopefully through this example you can see why we need to study quantum mechanics; without it, our picture of the world would be incomplete!

1.3 The Photoelectric Effect

Like the Planck radiation formula, the explanation of the photoelectric effect is also one that requires the assumption that energy is quantized. Famously, it was Einstein who proposed the correct mechanism for this phenomenon, which was ultimately the contribution that won him the Nobel prize, not his contributions of special and general relativity.

The photoelectric effect can be described as follows: when you shine light onto a metal surface, the light transfers energy onto the electrons trapped inside the metal, eventually transferring enough energy to allow them to "escape" the metal, resulting in a phenomenon called *photoemission*. Naturally, if light were to gradually transfer energy to these electrons, you should expect that even if the light were dim and low energy (e.g. microwaves or radio waves), the electrons would eventually accumulate enough energy to escape the metal, but in reality it does not — not all wavelengths of light generate photoemission, no matter the intensity and duration of exposure. By contrast, it seems that when the light reaches a certain frequency, photoemission *does* occur, even at low intensities and exposures. This frequency dependence is difficult to explain classically, under the assumption that any kind of light would transfer energy to the electrons.

So how do we explain this phenomenon? Einstein put forward the postulate that instead of treating light as a continuous beam of energy but instead **quantizing** it, this phenomenon can be explained. Specifically, we can imagine light as a stream of particles (called **quanta**), each carrying an energy $E = hf$ which it can impart to the electrons in the metal. He then theorized that there must be a minimum amount of energy required to remove an electron from the metal, called the *work function* ϕ. Since a photon can only deliver energy hf to the electron, this solves the frequency-dependent effect we couldn't explain from earlier. This also explains why changing the intensity nor duration matter: the energy of the light quanta is only frequency dependent, hence varying things like the intensity or duration have no bearing on photoemission.

1.4 The Double Slit Experiment

The double slit experiment is another famous phenomenon discovered in the early 19th century that challenged our classical view of light. The experiment is as follows: it involves taking a beam of light (usually a laser beam) shining upon a plate that has two parallel slits cut into them. Then, the light passing through the slits is captured on a screen behind the plate.

If light were to behave purely classically, you would expect to see two bright spots that trace out the shape of the slits on the screen. However, what is actually observed is an *interference pattern*: patches of light and dark spots spread across the screen that that gradually get fainter toward the edges. Now, if we are to assume that light is a beam of particles like we did with the photoelectric effect, this result makes no sense – only particles that fly through the slit make it onto the screen, and there is no force that allows them to change momentum so an interference pattern does not make sense.

However, this phenomenon can be explained if we take a different approach, and assume that light consists of waves instead of particles. This way, as waves travel through the slit, they experience a phenomenon known as *diffraction*, where light passing through the slits generate wavefronts that interfere with each other. This way, light going through one slit can interfere with the other, producing the interference pattern we see on the screen.

The double slit phenomenon, combined with the photoelectric effect, showcase a concept in quantum physics called the *wave-particle duality*. Specifically, it refers to the fact that light behaves both as a wave and a particle, depending on the context. In the double slit experiment, light behaves as a wave, whereas it behaves like a particle in the photoelectric effect.

It is often misunderstood that light somehow "chooses" to either be a wave or a particle depending on the experiment, but in reality wave-particle duality is better understood as light *simultaneously* behaving as a wave and a particle – doing the experiment simply allows us to see one of these two expressions.

1.5 The Case of the Electron

As a last example to motivate the need for quantum mechanics, there was a known problem with the classical model of the atom. In the late 19th and early 20th century, the leading model of the atom was the Bohr model, which depicted the nucleus with its protons and neutrons, and electrons orbiting around it. However, there was a problem with this model: by this point, it had already been established that moving charges radiate energy in the form of electromagnetic waves, and since the electron orbits around the nucleus you could calculate how much energy its orbit should radiate. As it turns out, if the electron really did orbit according to this model, then the electron would have crashed into the nucleus in 10^{-11} seconds!

Clearly this isn't true, and it really highlights why the classical models at the time were incomplete. Quantization resolves this however: by introducing *allowed* energies that the electron can exist in and removing the idea that the electron orbits around the nucleus, you resolve this problem immediately.

1.6 Conclusion

Hopefully through these examples you can see how inadequate our classical theories fail to explain the phenomena in the real world. Things like the photoelectric effect, the ultraviolet catastrophe and the electron radiation cannot be explained without introducing the quantization of energy, and phenomena like the double-slit experiment challenge our intuition, forcing us to accept that there are particles that can be both a wave and a particle at the same time. While this doesn't immediately motivate quantum behavior, it does illustrate that objects we once thought of as particles have wave-like properties as well.

Now that you've gotten an appreciation for the importance of quantum mechanics, let's now move on to develop the formal mathematics of quantum mechanics.

2 Mathematical Formalism

In the previous chapter, we gave a brief overview of the origins and development of quantum mechanics. This was aimed largely to provide meaningful historical context for how we ended up with such a theory. With that context clarified, we are now ready to explore the subject matter itself.

Of course, before we delve into our study of quantum mechanics, we will need to establish the fundamental underlying framework behind the subject. This will involve introducing a lot of relevant mathematical formalism. The formalism will focus around important elements of linear algebra, notably studying spaces of wave functions (*Hilbert spaces*). We will be introducing other mathematical principles later on as needed, but we summarize the key aspects of the relevant linear algebra here to give us a solid springboard.

There will be a *lot* of math in the following chapter. This chapter alone will introduce more math than any other individual chapter. This may feel rather overwhelming. Do not fret; the goal is not to master all of the math we're about to explain with the utmost rigor; rather, the goal is to obtain a basic framework so the math we'll use to develop the remaining theory of quantum mechanics will make sense.

Let's now begin with the formalism. For the time being, we will assume some prior familiarity working with linear algebra objects like vector spaces and linear maps, and a bit of knowledge about complex numbers, such as the definition of a complex conjugate. If you're interested in learning more about these, feel free to check out the appendices at the end of this book, or consult some linear algebra textbooks such as Axler's *Linear Algebra Done Right*.[1]. Now, on with the show!

2.1 The Wavefunction

The central focus of our analysis is the *wavefunction* of a system, which we'll denote with a capital or lowercase ψ. As the name suggests, this is a function that will take certain inputs (such as position, momentum, etc) and can be used to ascertain other properties (such as energy). Solving for the wavefunction of a system is the central focus of quantum mechanics,

[1]We highly encourage you to give this book a read! It covers all the formalism with a focus on rigor (we will also try to be rigorous, but only to the degree that it serves to make our point), and while it is not an explicitly physics-centric text, is a foundational text that will be expected of you in higher level theoretical physics.

since the wavefunction itself fully represents a system/state; this will be expanded on further in [POSTULATES REF].

Now, since the wavefunction has to represent physically realizable states (eg, an electron or a Hydrogen atom), there are certain restrictions we must enforce on what type of function this can be. What restriction can we put on a function describing some sort of physical system? Answering this question involves understanding what the wavefunction is meant to encode in the first place. The key insight is recognizing that we can make use of *probability* for this. In other words, when measuring a particle's position, we know that it **must** exist *somewhere* in space. Mathematically, this means that the probability of finding the wavefunction across all of space must be 1:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1$$

When ψ satisfies this property, we say that it is *normalized*. To make our lives a little bit simpler for now, we can loosen this restriction and only enforce that the function is *normalizable*:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx < \infty$$

This condition ends up being sufficient for our discussion, since any wavefunction satisfying this can be normalized by multiplying ψ by the reciprocal of the square root of the integral:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = A \implies \int_{-\infty}^{\infty} \left| \frac{1}{\sqrt{A}} \psi(x) \right|^2 \, dx = 1$$

Remark. The interpretation that the squared norm of the wavefunction encodes probability was a nontrivial leap that took physicists years to uncover. We applaud Max Born's discovery, but will make no attempt to naturally arrive at this conclusion ourselves.

The requirement that ψ is normalizable and it being complex valued are the only restrictions on the kinds of ψ we are allowed to consider. The collection of such ψ forms a vector space, and since ψ is complex-valued, we call this a *complex vector space*. We can then take this space, and attach an operation called the *inner product* to produce a *Hilbert space*.[2] Specifically, the inner product we use is:

$$\langle \psi(x), \phi(x) \rangle = \int_{-\infty}^{\infty} \psi^*(x) \phi(x) \, dx$$

where ψ, ϕ are two wavefunctions and the $*$ represents the complex conjugate. Combining this inner product with the set of ψ's defines our Hilbert space. The reason we do this is because, as we will see, this framework ends up being a very natural (and elegant) way to talk about wavefunctions and the relationships between them.

[2]A vector space equipped with a properly-defined inner product is called an *inner product space*. For a Hilbert space specifically, we enforce extra conditions on this inner product, but these won't concern us for a while.

2.1.1 Different Types of Wavefunctions

So far, we mainly considered wavefunctions defined as functions of position $\psi(x)$, but it's entirely possible we could also care about wavefunctions as functions of momentum (or any other variable). In the particular case of momentum, we can convert between the two using what's known as a *Fourier transform*:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int \psi(x) e^{-ipx/\hbar} \, dx$$

Unlike the positional wavefunctions $\psi(x)$, which live in what we call *configuration space*, the Fourier-transformed momentum wavefunctions $\phi(p)$ live in *momentum space*. You may question (and rightfully so) whether these wavefunctions are similar enough to be comparable. Fortunately, their norms in their respective spaces are equal (thanks to the *Parseval identity*, which we will not prove):

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = \int_{-\infty}^{\infty} |\phi(p)|^2 \, dp$$

Thus, these wavefunctions in momentum space form their own separate Hilbert space (in terms of the momentum p, instead of the position x). It should be stressed that there is no one *correct* wavefunction: $\psi(x)$ is just as valid of a wavefunction as $\phi(p)$, and the choice of which one to use will depend on the problem at hand. Given that we have these choices though, this is a good point to take a step back and develop some general notation that is agnostic to the variable we ultimately choose.

2.2 Dirac Notation (Bras, Kets, etc.)

Most of our study will revolve around analyzing quantum states, which begs an important question: which basis should we choose? How do we properly represent states?

Earlier, we noted how there are various Hilbert spaces of wavefunctions that may be of interest to us (there are even more, but we won't cover all of them). So, how do we denote a state without giving undue prejudice to one representation over another? This is where we want to establish some notational formalism that abstracts away from prejudicial state representations. Thus, we want to discuss *Dirac notation*.

2.2.1 Vector Representations

To represent a state of a given system, we will use a *ket vector*, denoted $|\rangle$. So, instead of writing our state as a wavefunction of a particular variable, like $\psi(x)$, we will abstract to the more general state $|\psi\rangle$.

Remark. When solving physical problems, we will often need to decide on a space and revisit the traditional wavefunction as a function of a variable, but in the most general case, we will use ket vectors. In fact, we use this more abstract representation to later derive the more traditional wavefunctions.

To motivate why we want to do this: recall that in the last section we defined the set of allowed ψ as a vector space. So, it's only natural that we think of each ψ as a vector and represent it using $|\psi\rangle$. Since these are vectors, we expect them to behave as such, satisfying properties like:

$$c\,|\psi\rangle = |\psi\rangle\, c$$

for any scalar c. The complex vector space in which these $|\psi\rangle$ live we will denote as $\mathcal{E}$.

2.2.2 Dual Spaces

Linear algebra also tells us that vectors and their accompanying vector space have corresponding *dual vectors* that live in the *dual space*. These are our *bras* $\langle|$, which live in $\mathcal{E}^*$. Put very simply, these dual vectors (otherwise known as *covectors* or *forms*), describe linear operators that act on vectors that live in $\mathcal{E}$:

$$\langle\psi| : \mathcal{E} \to \mathbb{C}$$

In other words, any linear operation that takes ket vectors as an input and spits out a complex number will belong to our dual space. So, we can act a bra on a corresponding ket:

$$\langle\alpha|\,(|\beta\rangle) = \langle\alpha|\beta\rangle$$

For simplicity, we can drop the parentheses in these products. Given how the dual vector is defined, we then know that this quantity $\langle\alpha|\beta\rangle$ will be a complex number.

When we want to emphasize that a bra is a complex-valued operator that takes in a ket as an input, we may reintroduce parentheses to accentuate this. Otherwise, we will drop it.

Bras are defined as linear operators, meaning we naturally expect them to satisfy the relevant *linearity* and *homogeneity* conditions:

$$\langle\alpha|\,(c_1\,|\psi_1\rangle + c_2\,|\psi_2\rangle) = c_1\,\langle\alpha|\psi_1\rangle + c_2\,\langle\alpha|\psi_2\rangle$$

Additionally, scalar products and sums work exactly as we would expect:

$$(c\,\langle\alpha|)(|\psi\rangle) = c\,\langle\alpha|\psi\rangle\,, \quad (\langle\alpha_1| + \langle\alpha_2|)(|\psi\rangle) = \langle\alpha_1|\psi\rangle + \langle\alpha_2|\psi\rangle$$

Remark. If you're struggling to remember which name refers to which state, think about bras and kets coming in pairs, together forming "brackets" $\langle|\rangle$. the "bra" comes first and points left, thereby representing the dual, whereas the "ket" comes second and points right, representing the traditional vector.

2.2.3 Connecting Bras and Kets

Even though we established what bras and kets are and the correspondence between the two may seem obvious, that's not good enough for us mathematically: we need an explicit way to convert between bras and kets. For this, let's consider some function g that takes in two kets and spits out a complex number:

$$g : \mathcal{E} \times \mathcal{E} \to \mathbb{C}$$

This function is called a *metric* (or *scalar product*) on $\mathcal{E}$. We want this function g to satisfy a few key properties:

1. **Linearity in second operand**: $g(|\psi\rangle, c_1 |\phi_1\rangle + c_2 |\phi_2\rangle) = c_1 g(|\psi\rangle, |\phi_1\rangle) + c_2 g(|\psi\rangle, |\phi_2\rangle)$
2. **Antilinearity in first operand**: $g(c_1 |\psi_1\rangle + c_2 |\psi_2\rangle, |\phi\rangle) = c_1^* g(|\psi_1\rangle, |\phi\rangle) + c_2^* g(|\psi_2\rangle, |\phi\rangle)$
3. **Symmetry** (or *Hermiticity*): $g(|\psi\rangle, |\phi\rangle) = g(|\phi\rangle, |\psi\rangle)^*$
4. **Positive-Definite**: $g(|\psi\rangle, |\psi\rangle) \geq 0$; equal only when $|\psi\rangle = 0$

Here, the * refers to the *complex conjugate* of the number:

$$c = a + bi \implies c^* = a - bi$$

With that metric established, we are able to properly define the *dual correspondence* that will convert between kets and bras. We'll denote this correspondence $D : \mathcal{E} \to \mathcal{E}^*$. This way, when we write a ket $|\psi\rangle$, we know that the associated bra under this dual correspondence will be denoted $\langle\psi|$. Given this metric, we can then explicitly define the bra $\langle\psi|$ in terms of how it acts on any random ket $|\phi\rangle$:

$$\langle\psi| (|\phi\rangle) = g(|\psi\rangle, |\phi\rangle)$$

We'll denote the bra we get from a ket's dual correspondence through a *dagger* †:

$$\langle\psi| = (|\psi\rangle)^\dagger, \quad |\psi\rangle = (\langle\psi|)^\dagger$$

This dagger means then that $\langle\psi|$ is the *Hermitian conjugate* of $|\psi\rangle$ (and vice-versa).

Given how we defined our scalar product to be antilinear in the first operand, we expect the dual correspondence to also be antilinear:

$$(c_1 |\psi_1\rangle + c_2 |\psi_2\rangle)^\dagger = c_1^* \langle\psi_1| + c_2^* \langle\psi_2|$$

Given how we expressed this correspondence, we naturally expect that repeating the operation necessarily returns us to the original state:

$$(|\psi\rangle)^{\dagger\dagger} = |\psi\rangle, \quad (\langle\psi|)^{\dagger\dagger} = \langle\psi|$$

With all of this clarified, we can proceed to simplify our notation of bra's actions on kets:

$$\langle\psi| (|\phi\rangle) = g(|\psi\rangle, |\phi\rangle) = \langle\psi|\phi\rangle$$

This means we can also rewrite our Hermiticity and positive-definite conditions:

$$\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*, \quad \langle\psi|\psi\rangle \geq 0$$

This has allowed us to completely rid ourselves of the g-function notation, meaning we can now deal exclusively with bras and kets. This operation of acting a bra on a ket has another name which may be familiar to those who have studied linear algebra: the *inner product*.

Remark. You may ask why we wrote a $*$ over the combined bra-ket product above instead of a dagger. This is because we will typically reserve the dagger notation for Hermitian conjugates of vectors and operators, while the star refers to the complex conjugate of a scalar (number). Since a bra acted on a ket necessarily produces a scalar, its conjugate will also necessarily be a number.

2.2.4 Some Practice: The Schwarz Inequality

Let's put our newly-developed notation to some use and prove a very important theorem in linear algebra called the *Schwarz inequality*.

Theorem 2.1 (Schwarz Inequality)**.**

$$|\langle\psi|\phi\rangle|^2 \leq \langle\psi|\psi\rangle\,\langle\phi|\phi\rangle \quad \forall\,|\psi\rangle, |\phi\rangle$$

Equality holds iff[a] $|\psi\rangle$ *and* $|\phi\rangle$ *are* linearly dependent *(ie,* $|\psi\rangle = c\,|\phi\rangle$ *for some scalar* c*).*

[a] This is a notational shorthand that means "if and only if".

Proof. Set some new ket $|\alpha\rangle = |\psi\rangle + \lambda\,|\phi\rangle$, where $\lambda \in \mathbb{C}$, and take the inner product with itself:

$$\langle\alpha|\alpha\rangle = (\langle\psi| + \lambda^*\langle\phi|)(|\psi\rangle + \lambda\,|\phi\rangle) = \langle\psi|\psi\rangle + \lambda\,\langle\psi|\phi\rangle + \lambda^*\,\langle\phi|\psi\rangle + |\lambda|^2\,\langle\phi|\phi\rangle \geq 0$$

We know by definition that $\langle\alpha|\alpha\rangle \geq 0$ (property 4 of the metric g), with equality only when $|\alpha\rangle = 0$, or equivalently when $|\psi\rangle = -\lambda\,|\phi\rangle$, which is precisely our linear dependence condition. We'll skip the trivial case where $|\phi\rangle = 0$, and address the more interesting case when $|\phi\rangle \neq 0$. Since the above inequality holds for all $|\psi\rangle$, $|\phi\rangle$ and λ, let's set $\lambda = -\frac{\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle}$:

$$\begin{aligned}
&\langle\psi|\psi\rangle - \frac{\langle\phi|\psi\rangle\,\langle\psi|\phi\rangle}{\langle\phi|\phi\rangle} - \frac{\langle\psi|\phi\rangle\,\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle} + \frac{\langle\psi|\phi\rangle\,\langle\phi|\psi\rangle\,\langle\phi|\phi\rangle}{\langle\phi|\phi\rangle^2} \\
&\quad = \langle\psi|\psi\rangle - 2\frac{\langle\psi|\phi\rangle\,\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle} + \frac{\langle\psi|\phi\rangle\,\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle} = \langle\psi|\psi\rangle - \frac{\langle\psi|\phi\rangle\,\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle} = \langle\psi|\psi\rangle - \frac{|\langle\psi|\phi\rangle|^2}{\langle\phi|\phi\rangle} \geq 0
\end{aligned}$$

Moving the second term to the right and multiplying by $\langle\phi|\phi\rangle$ gives us our original inequality, as desired. ■ □

As we can see, Dirac notation is compact, consistent, and comprehensible for problem-solving.[3]

Remark. We illustrated this mathematical proof outside of an appendix to better demonstrate the power of Dirac notation, as well as to illustrate an important theorem that we'll make heavy use of later on (notably with proving the *Heisenberg uncertainty relations*).

2.2.5 Why Bother with Dirac Notation?

This is all well and good, but why are we bothering with Dirac notation in the first place? After all, many introductory courses in quantum mechanics simply use traditional wavefunction notation and don't make use of this notation until *much* later, so why are we bringing it up now?

There are several reasons for why we want to introduce this convention sooner rather than later. Chief among them is the notion of *generality*. Starting with more special cases like configuration and momentum-space representations and then trying to work backwards to generalizations later can often confuse the matter and make the abstraction seem pointless or needlessly convoluted. Starting with a more general formalism and then deriving special cases later can instead give us better intuition on the generalizations. This also allows us to be more rigorous in our development of the material, as everything will come naturally from this formalism, rather than forcing us to pull certain things out of a hat with minimal explanation as to how we got there.

This does come at a cost, though: it requires us to be much more mathematically technical at the start. We have already delved pretty deep into certain fundamental linear algebra concepts to make any of this make sense. Do not worry; with just a little bit of background, this will make our lives much easier in future discussions.

Soon, we will derive our usual wavefunction representations from Dirac notation, which will make things slightly easier to understand. In the meantime, do not fret; we have done all of this to make our foundation strong.

2.3 Operators and Commutators

Let's now pivot our attention to *operators*, which are how we'll extract meaningful information from our wavefunctions. For our purposes in nonrelativistic quantum mechanics, we will primarily be interested in two types of operators: *linear* operators and *antilinear* operators. These are both mappings from a ket space $\mathcal{E}$ onto itself (i.e. $A : \mathcal{E} \to \mathcal{E}$), but they behave slightly differently on linear combinations of kets:

[3] You may think initially that this proof isn't very comprehensible, but think about the alternative: if we didn't use Dirac notation, we'd have to write every inner product as an integral, every ψ and ϕ as $\psi(x)$ and $\phi(x)$, which would make the expression at least twice as long, and much more intimidating.

$$
\begin{aligned}
A(c_1 |\psi_1\rangle + c_2 |\psi_2\rangle) &= c_1 A |\psi_1\rangle + c_2 A |\psi_2\rangle && \text{(linear)} \\
A(c_1 |\psi_1\rangle + c_2 |\psi_2\rangle) &= c_1^* A |\psi_1\rangle + c_2^* A |\psi_2\rangle && \text{(antilinear)}
\end{aligned}
$$

In the spirit of total transparency, we will almost exclusively deal with *linear* operators, but mention antilinear operators as well because we will analyze one such case much later on.

Remark. The keen among you may recognize this one antilinear operator of interest as the *time reversal* operator. Antilinear operators will appear more frequently when we transition to relativistic quantum mechanics and quantum field theory, but we won't do that for a very long time.

Given how operators are defined as a mapping from a ket space onto itself, we know that these operators will take in ket vectors as an input and then spit out some other ket vector as an output.

2.3.1 Commutators

From linear algebra, we know that these operators can be scaled by complex numbers and added to each other to produce new operators on the same ket spaces, meaning they also form their own vector space. What about multiplication? Well, we expect the multiplication to be associative for any combination of linear operators A, B, and C:

$$A(BC) = (AB)C$$

However, it won't necessarily be commutative:

$$AB \neq BA$$

Why not? Let's illustrate with a real-world example.

Example 2.1 (Taking a Stroll)**.** Imagine you are navigating some space and start out facing east. If you walk forward, then turn left, you'll end up further east than when you started, facing north. What would happen if you instead turned left first, then walked forward? Turning left when facing east would make you now face north, meaning that you'd end up north of where you started, facing north. Despite only performing two steps, the order clearly matters.

[INSERT TIKZ HERE]

If you think of operators acting on ket vectors as instructions we give someone to perform, we see that swapping the order of the operators is equivalent to giving instructions in a different order, which can produce completely different results:

$$
\begin{aligned}
AB\,|\psi\rangle &= A(B\,|\psi\rangle) = A(B(|\psi\rangle)) = A(|\phi_B\rangle) = |\alpha\rangle \\
BA\,|\psi\rangle &= B(A\,|\psi\rangle) = B(A(|\psi\rangle)) = B(|\phi_A\rangle) = |\beta\rangle
\end{aligned}
$$

Thus, multiplication of operators will not act the same as standard scalar multiplication, meaning we'll have to be careful when specifying order.

Keeping in mind that operators will not necessarily commute, we need a way to quantify exactly how close (or far) two operators are from commuting with each other. To help us with this, we define the *commutator*:

Definition 2.1 (Commutator)**.** The *commutator* of two operators A and B is the difference between their ordered products:

$$[A, B] = AB - BA$$

Similarly, we define the *anticommutator*:

Definition 2.2 (Anticommutator)**.** The *anticommutator* of two operators is the sum of their ordered products:

$$\{A, B\} = AB + BA$$

Naturally, the operators will commute when their commutator is 0; in that case, we can freely interchange their order. Some operators may have an associated *inverse*, which we'll denote A^{-1}. This inverse "undoes" the action of the operator:

$$AA^{-1} = A^{-1}A = 1$$

Above, the "1" denotes the *identity matrix*, rather than a number. In some instances, we will opt to write I for the identity, but 1 is also acceptable.

Notice the hesitant phrasing of *if* the inverse exists. While it's natural to assume that every operation can somehow be undone, this is unfortunately not always the case. Certain conditions must be met for an operator to have a valid inverse, which are addressed in more detail in our Linear Algebra appendix and other materials. Fortunately for us, many key operators will in fact be invertible.

When all available inverses exist, we have the following relation for inverting products of operators:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Remark. Another important consideration we glossed over here is the *dimensionality* of our vector spaces. As it turns out, finite and infinite-dimensional vector spaces behave differently, meaning there's extra work to be done when trying to extend properties that work in finite-dimensional vector spaces to infinite-dimensional ones. One example is the notion of inverses. When our vector space is finite-dimensional, the inverse of an operator will be both a *left-inverse* and a *right-inverse*, meaning it will undo the operation regardless of whether we act it before or after the operator. This intuitively makes sense, since the inverse of an operation shouldn't depend on order. However, in infinite-dimensional spaces, it's entirely possible (for reasons which we'll not explore) for an operator to possess *only* a left or right-inverse, but not necessarily both. This, among many other reasons, is why we want to be very careful when making the jump from finite to infinite-dimensional spaces (and relevant operators/kets).

A few basic properties arise purely from the definition of the commutator:

1. **Linearity in first operand**: $[c_1 A_1 + c_2 A_2, B] = c_1[A_1, B] + c_2[A_2, B]$
2. **Linearity in second operand**: $[A, c_1 B_1 + c_2 B_2] = c_1[A, B_1] + c_2[A, B_2]$
3. **Antisymmetry**: $[A, B] = -[B, A]$

There are two additional properties the commutator obeys that we want to address separately. First is the *Jacobi identity*:

Lemma 2.1 (Jacobi Identity)**.**

$$[A,[B,C]] + [B,[C,A]] + [C,[A,B]] = 0$$

Proof. Expand out each nested commutator:

$$\begin{aligned}
[A,[B,C]] &= [A,(BC-CB)] = A(BC-CB) - (BC-CB)A = \\
&= ABC - ACB - BCA + CBA \\
[B,[C,A]] &= [B,(CA-AC)] = B(CA-AC) - (CA-AC)B = \\
&= BCA - BAC - CAB + ACB \\
[C,[A,B]] &= [C,(AB-BA)] = C(AB-BA) - (AB-BA)C = \\
&= CAB - CBA - ABC + BAC
\end{aligned}$$

Now, add these expressions together and cancel negative terms:

$$\begin{aligned}
[A,[B,C]] + [B,[C,A]] + [C,[A,B]] = &\cancel{ABC} - \cancel{ACB} - \cancel{BCA} + \cancel{CBA} + \\
&+ \cancel{BCA} - \cancel{BAC} - \cancel{CAB} + \cancel{ACB} + \\
&+ \cancel{CAB} - \cancel{CBA} - \cancel{ABC} + \cancel{BAC} = \boxed{0} \qquad \checkmark
\end{aligned}$$

As desired. ■ □

One more set of basic properties the commutator satisfies are collectively called the *derivation property* (or *Leibniz rule*):

Lemma 2.2 (Leibniz Rule)**.**

$$[AB, C] = A[B, C] + [A, C]B, \quad [A, BC] = B[A, C] + [A, B]C$$

Proof. We'll prove both of these by expanding the left-hand and right-hand sides to show they're equal, starting with the first:

$$\begin{aligned}
[AB, C] &= (AB)C - C(AB) = ABC - CAB = \\
&= ABC - CAB + (ACB - ACB) = \\
&= ABC - ACB + ACB - CAB = \\
&= A(BC - CB) + (AC - CA)B = A[B, C] + [A, C]B \quad \checkmark
\end{aligned}$$

Here, we cleverly added 0 to avoid changing the value of the LHS, while also giving us extra terms to rewrite the RHS. We do the same for the second expression:

$$\begin{aligned}
[A, BC] &= A(BC) - (BC)A = ABC - BCA = \\
&= ABC - BCA + (BAC - BAC) = \\
&= BAC - BCA + ABC - BAC = \\
&= B(AC - CA) + (AB - BA)C = B[A, C] + [A, B]C \quad \checkmark
\end{aligned}$$

As desired. ■ □

We will deal with commutators a lot moving forward, and we will often encounter very long and messy expressions that we would greatly prefer to simplify. Some of these properties will help us achieve that. We will, of course, introduce more commutator relations in the future, but we have to start somewhere.

2.3.2 Operators on Dual Vectors

Given how we have defined operators, we expect them to act on a ket vector and output some new ket vector:

$$A \left|\psi\right\rangle = \left|\phi\right\rangle$$

In other words, this operator will perform some kind of operation on the ket vector and modify it. What about bras and dual vectors? What kind of action will an operator perform in that scenario?

We will use the following notation to denote the action of an operator on a bra:

$$\left\langle\psi\right| A = ?$$

Colloquially, we will say that operators act on kets to the right and bras to the left.

Since operators acting on kets will produce other kets, we should expect operators acting on bras to produce other bras:

$$\langle\psi| A = \langle\phi|$$

Given how we defined operators and how bras and kets are related, it is not inconveivable for us to extend their definition to behave as such.[4] Now, given this, we can expect to envision an operator sandwiched between a bra and a ket as either a resultant bra acting on ket or vice versa:

$$(\langle\psi| A) |\phi\rangle = \langle\psi| (A |\phi\rangle)$$

This means we can drop the parentheses and treat this as one unit:

$$(\langle\psi| A)(|\phi\rangle) = \langle\psi|A|\phi\rangle$$

We call this combined expression the *matrix element*.

> *Remark.* In the context of linear algebra and linear maps, the term "matrix element" begins to make sense. Since a matrix is fundamentally a representation of a linear transformation's action on a given vector input, sandwiching an operator between a bra and a ket can be interpreted as an extraction of the relevant matrix element of that operator.

2.3.3 Outer Products

We previously defined the inner product as a bra acting on a ket from the left, which would produce a scalar by the definitions of bras and kets, respectively. What if we instead have a bra acting on a ket from the right? Let's act this new thing on a ket and see what we should expect:

$$(|\alpha\rangle \langle\beta|) |\psi\rangle = |\alpha\rangle (\langle\beta|\psi\rangle) = |\alpha\rangle c_1 = c_1 |\alpha\rangle = |\gamma_1\rangle$$

Thus, this new combination, which we'll call the *outer product*, should produce a resultant ket when acted on a ket. Similarly, when acted on a bra, we should expect a resultant bra:

$$\langle\psi| (|\alpha\rangle \langle\beta|) = \langle\psi|\alpha\rangle \langle\beta| = c_2 \langle\beta| = \langle\gamma_2|$$

Thus, this outer product must itself be a special type of linear operator. Note that the bra action is not the direct definition of the outer product; it is simply a consequence of the ket action, which itself is the proper definition.

[4]Technically, one must distinguish between the operator that sends kets to kets from the dual operator A' that sends bras to bras. However, here we can *extend* the definition of an operator to act on bras as $\langle\psi| A = \langle\phi|$ when $A |\psi\rangle = |\phi\rangle$, without clashing with our original definition of A, and as such we can use the same symbol A for the map and its dual.

2.3.4 Bases

Now, if we have a vector space, we should have a way to build this space out of a few fundamental vectors. A set of linearly independent vectors that can be used in some combination to produce any arbitrary vector in the space is called a *basis* for the vector space. We then use the length of this basis set to define the *dimension* of the vector space.

Remark. This is another key place where infinite-dimensional vector spaces behave differently from their finite-dimensional counterparts. We should expect such a space to also possess a basis, since we need to be able to construct it using some combination of vectors, but *how many* vectors should we have? We know from discrete mathematics that there are different *kinds* of infinities, namely **countable** and **uncountable** ones (put simply, countable infinities are ones where we can enumerate each element, and uncountable ones are those where it's impossible to do so). Fortunately for us, one of the properties of a Hilbert space is that it possesses a countably infinite basis, which is good for our purposes and the end of our discussion of the matter.

We will often require to decompose a vector space into an appropriate basis, or use a basis to construct a relevant vector space. Since the only requirements for a set to become a basis are for it to be the appropriate length, have each vector be linearly independent, and have the vectors span the entire space, it is not inconceivable to have multiple different bases generating a single vector space.

Let's label some discrete basis of some ket space with numbered kets: $\{|n\rangle, n = 1, 2, \dots\}$. If all vectors in the basis are *orthogonal* (i.e. $\langle n|m\rangle = 0$ when $n \neq m$), we call the basis an *orthogonal* basis. If, furthermore, all of the vectors of an orthogonal basis are normalized (each has magnitude 1), we call the basis *orthonormal*. To properly notate, let's introduce a new function which we'll see a lot later on called the *Kronecker delta*[5]:

Definition 2.3 (Kronecker Delta)**.** The *Kronecker delta* is defined like so:

$$\delta_{mn} = \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases}$$

Thus, we can rewrite the orthonormality condition for our basis into the following really compact form:

$$\langle n|m\rangle = \delta_{mn}$$

Remark. We will use *index notation* like the Kronecker delta above to simplify notation often throughout quantum mechanics (and in some instances in other fields of physics as well). Index notation will be clarified a bit more in some appendices, and will come more in

[5]This is technically a function, but not really the type of function we are used to dealing with. It is more helpful to think of it as a notational convention.

handy when we start dealing with tensors, which will come a little later.

Given any basis, we can represent any arbitrary ket vector $|\psi\rangle$ in that space as a linear combination of some these basis vectors:

$$|\psi\rangle = \sum_n c_n |n\rangle = \sum_n |n\rangle c_n$$

These expansion coefficients c_n are given by the following inner product if the basis is orthonormal:

$$c_n = \langle n|\psi\rangle$$

This is pretty straightforward to justify given the definition of orthonormal vectors:

$$\langle n|\psi\rangle = \langle n| \left(\sum_m |m\rangle c_m\right) = \delta_{nm} c_m = 0 + 0 + \cdots + 1 * c_n = c_n$$

This means we can rewrite our expression as:

$$|\psi\rangle = \sum_n \langle n| \langle n|\psi\rangle = \left(\sum_n \langle n|n\rangle\right) |\psi\rangle$$

Since $|\psi\rangle$ is an arbitrarily-chosen ket vector, we can remove it from our expression to get a *resolution of the identity*:

$$\boxed{1 = \sum_n |n\rangle \langle n|}$$

2.3.5 A Special Case: Hermitian and Unitary Operators

We will see many kinds of operators, but in quantum mechanics, we are particularly interested in two special types of operators.

Before we discuss them, we need to extend the notion of Hermitian conjugation from kets to operators. Given some operator $A : \mathcal{E} \to \mathcal{E}$, we expect the Hermitian conjugate $A^\dagger : \mathcal{E} \to \mathcal{E}$ to be another linear operator with the following action on an arbitrary ket:

$$A^\dagger |\psi\rangle = (\langle\psi| A)^\dagger$$

We have a couple of immediate consequences that result directly from this definition, which we compile here:

Corollary 2.1 (Properties of the Hermitian Conjugate)**.** *The following follow from the definition of the Hermitian conjugate of an operator:*

1. $\langle\phi|A^\dagger|\psi\rangle = \langle\psi|A|\phi\rangle^*$
2. $(A^\dagger)^\dagger = A$

3. $(c_1 A_1 + c_2 A_2)^\dagger = c_1^* A_1^\dagger + c_2^* A_2^\dagger$
4. $(AB)^\dagger = B^\dagger A^\dagger$
5. $(|\alpha\rangle \langle\beta|)^\dagger = |\beta\rangle \langle\alpha|$

Proof. The proofs for each are quite straightforward, but let's illustrate them here. Starting with the first:

$$\langle\phi|A^\dagger|\psi\rangle = \langle\phi| (A^\dagger |\psi\rangle) = \langle\phi| (\langle\psi| A)^\dagger = (\langle\psi| A)^\dagger (\phi)^\dagger = \langle\psi|A|\phi\rangle^*$$

Now, for the second, we can apply the above property twice:

$$\begin{aligned} \langle\psi|(A^\dagger)^\dagger|\phi\rangle &= \langle\phi|A^\dagger|\psi\rangle^* = (\langle\psi|A|\phi\rangle^*)^* = \langle\psi|A|\phi\rangle \\ \therefore (A^\dagger)^\dagger = A \quad \checkmark \end{aligned}$$

Next:

$$\begin{aligned} (c_1 A_1 + c_2 A_2)^\dagger |\psi\rangle &= (\langle\psi| (c_1 A_1 + c_2 A_2))^\dagger = ((\langle\psi| c_1 A_1) + (\langle\psi| c_2 A_2))^\dagger = \\ &= (\langle\psi| c_1 A_1)^\dagger + (\langle\psi| c_2 A_2)^\dagger = A_1^\dagger (c_1^* |\psi\rangle) + A_2^\dagger (c_2^* |\psi\rangle) = \\ &= c_1^* A_1^\dagger |\psi\rangle + c_2^* A_2^\dagger |\psi\rangle = (c_1^* A_1^\dagger + c_2^* A_2^\dagger) |\psi\rangle \\ \therefore (c_1 A_1 + c_2 A_2)^\dagger = c_1^* A_1^\dagger + c_2^* A_2^\dagger \quad \checkmark \end{aligned}$$

For the fourth, we use what we proved in the first:

$$\begin{aligned} \langle\phi|(AB)^\dagger|\psi\rangle &= \langle\psi|AB|\phi\rangle^* = (((\langle\psi| A)(B |\phi\rangle))^* = (((\langle\psi| A)^\dagger (B |\phi\rangle)^\dagger) \\ &= (\langle\phi| B^\dagger)(A^\dagger |\psi\rangle) = \langle\phi|B^\dagger A^\dagger|\psi\rangle \\ \therefore (AB)^\dagger = B^\dagger A^\dagger \end{aligned}$$

Finally, for the last one, we directly invoke the previous part (as well as dual correspondence between bras and kets):

$$(|\alpha\rangle \langle\beta|)^\dagger = (\langle\beta|)^\dagger (|\alpha\rangle)^\dagger = |\beta\rangle \langle\alpha| \quad \checkmark$$

We have shown all properties are true, as desired. ■ □

Now, we have properly defined the action of Hermitian conjugation on bras, kets, and operators. As a side note, it's helpful to describe the action of Hermitian conjugation on scalars (numbers) as simply complex conjugation:

$$c^\dagger = c^*$$

With this in mind, we can combine all of these together and write that the Hermitian conjugate simply reverses the order of the quantities and swaps them out for their Hermitian conjugate:

$$\boxed{(c_1 A_1 |\psi_1\rangle + \langle\psi_2| c_2 A_2)^\dagger = \langle\psi_1| A_1^\dagger c_1^* + c_2^* A_2^\dagger |\psi_2\rangle}$$

We are now finally ready to discuss a special class of operators that we will see a lot of in quantum mechanics. We start with *Hermitian* operators:

Definition 2.4 (Hermitian Operator)**.** An operator A is *Hermitian* if it is equal to its own Hermitian conjugate:

$$A^\dagger = A$$

Using the first auxiliary property we proved that followed from the definition, we can equivalently write this condition in terms of matrix elements:

$$A \text{ Hermitian} \implies \langle\psi|A|\phi\rangle = \langle\phi|A|\psi\rangle^* \quad \forall\, |\psi\rangle\,, |\phi\rangle$$

Similarly, we have *anti-Hermitian* operators:

Definition 2.5 (Anti-Hermitian Operator)**.** An operator A is *anti-Hermitian* if it is the additive inverse of its Hermitian conjugate:

$$A^\dagger = -A$$

An operator being Hermitian or anti-Hermitian is a special case; in other words, if an operator isn't Hermitian, it's not necessarily guaranteed to be anti-Hermitian. In general, operators will be neither. That being said, it is always possible to uniquely decompose any operator A into a sum of a Hermitian and an anti-Hermitian operator.

Lemma 2.3 (Hermitian Decomposition of Operators)**.** *Any operator A can be uniquely decomposed into the sum of a Hermitian and anti-Hermitian operator:*

$$A = \frac{A + A^\dagger}{2} + \frac{A - A^\dagger}{2}$$

Proof. The decomposition itself is trivial, so there's no need to prove it. The more interesting part to prove is that these newly-generated operators are Hermitian and anti-Hermitian. We prove by taking their respective Hermitian conjugates:

$$\left(\frac{A + A^\dagger}{2}\right)^\dagger = \frac{A^\dagger + A^{\dagger\dagger}}{2} = \frac{A^\dagger + A}{2} = \frac{A + A^\dagger}{2} \qquad \text{(Hermitian)}$$

$$\left(\frac{A + A^\dagger}{2}\right)^\dagger = \frac{A^\dagger - A^{\dagger\dagger}}{2} = \frac{A^\dagger - A}{2} = -\left(\frac{A - A^\dagger}{2}\right) \qquad \text{(anti-Hermitian)}$$

Thus, the first term in the decomposition is a Hermitian operator, and the second is anti-Hermitian. Since the decomposition is possible to write for any operator A, we can always uniquely decompose into a sum of Hermitian and anti-Hermitian operators, as desired. ■ □

We will call a Hermitian operator A *positive-definite* if the following is true for all kets $|\psi\rangle$:

$$\langle\psi|A|\psi\rangle > 0$$

If we loosen the restriction and allow for this element to be 0 by swapping for $\geq$ in the inequality, we call that type of Hermitian operator *positive semidefinite.*

One more important theorem on Hermitian operators:

Theorem 2.2 (Product of Hermitian Operators)**.** *The product of two Hermitian operators A, B is Hermitian if and only if A and B commute.*

Proof. The proof is quite simple, but because we have a biconditional, we must prove both directions.

($\Longrightarrow$) Starting with the forward direction, we will prove that if the product of two Hermitian operators is Hermitian, then they must commute. If the product of A and B is Hermitian, then we have:

$$(AB)^\dagger = B^\dagger A^\dagger = (B)(A) = AB$$

This last equality implies that they commute:

$$[A, B] = AB - BA = 0 \quad \checkmark$$

($\Longleftarrow$) Now, we prove the reverse direction, showing that if two Hermitian operators commute, their product must be Hermitian. If A and B commute, then we have:

$$[A, B] = AB - BA = 0 \implies AB = BA$$

Taking the Hermitian conjugate of AB shows us that the product is Hermitian:

$$(AB)^\dagger = B^\dagger A^\dagger = (B)(A) = BA = AB \quad \checkmark$$

Having shown both directions, we are done. ■ □

Now, we move on to *unitary* operators.

Definition 2.6 (Unitary Operator)**.** An operator U is *unitary* if its Hermitian conjugate is equal to its inverse:

$$UU^\dagger = U^\dagger U = 1 \implies U^{-1} = U^\dagger$$

This definition has one immediate (and important) consequence:

Corollary 2.2 (Product of Unitary Operators)**.** *The product of two unitary operators U_1, U_2 will always be unitary.*

Proof. The proof is quite simple. Let's throw their product into the definition:

$$\begin{aligned}(U_1U_2)(U_1U_2)^\dagger &= U_1U_2(U_2^\dagger U_1^\dagger) = U_1U_2U_2^\dagger U_1^\dagger = U_1U_2U_2^{-1}U_1^{-1} = \\ &= U_1(U_2U_2^{-1})U_1^{-1} = U_1(1)U_1^{-1} = U_1U_1^{-1} = 1 \\ (U_1U_2)^\dagger(U_1U_2) &= (U_2^\dagger U_1^\dagger)U_1U_2 = U_2^\dagger U_1^\dagger U_1U_2 = U_2^{-1}U_1^{-1}U_1U_2 = \\ &= U_2^{-1}(U_1^{-1}U_1)U_2 = U_2^{-1}(1)U_2 = U_2^{-1}U_2 = 1\end{aligned}$$

Thus, we have shown that

$$(U_1U_2)(U_1U_2)^\dagger = (U_1U_2)^\dagger(U_1U_2) = 1,$$

showing that the product of two unitary operators is unitary, as desired. ■ □

2.3.6 Eigenvalues and Eigenkets

Operators will act on kets to produce other kets (and act on bras to produce other bras). In many instances, this new resultant ket will be completely new and not entirely related to the original input ket. Sometimes, however, acting an operator on a certain ket will simply scale the original ket by some factor:

$$A\,|u\rangle = a\,|u\rangle$$

Here, the ket vector $|u\rangle$ is called an *eigenket* of A (or, equivalently in linear algebra, an *eigenvector* of the matrix represented by A), and the complex number a is the associated (right) *eigenvalue* of A[6].

We also expect that there may exist a special bra that will only be scaled under the operator:

$$\langle\nu|\,A = b\,\langle\nu|$$

In this case, we call the bra $\langle\nu|$ an *eigenbra* of A and b a (left) eigenvalue.

Remark. Notice that we labeled eigenvalues associated with eigenkets and eigenbras with the (right) and (left) prefixes, respectively. We do this, because we want to be as explicit as possible with whether the scaling factor is associated with the operator acting to the *right* on a ket or to the *left* on a bra. We will typically drop these prefixes when dealing with problems, because in finite dimensions, every right-eigenvalue will also be a left-eigenvalue (and vice versa), but this need not be the case in infinite dimensions.

Operators will typically have a bunch of associated eigenvalues. We call the set of all of these eigenvalues the *spectrum of the operator*. These spectra may be *discrete* (separate, individual points), *continuous* (an infinite line), or *mixed* (some combination of discrete and continuous). We will see examples of each of these as we develop more quantum mechanics.

[6]The terms "eigenvalue" and "eigenvector" are built from the German *eigen*, which means "self".

Generally, there won't really be a simple way to relate different eigenkets and eigenbras for a given operator; we will simply solve for them and make use of them. However, when we're dealing with *Hermitian* operators specifically, we will get a few simplifications that actually will allow us to relate them.

We start with the first meaningful simplification:

Lemma 2.4 (Eigenvalues of Hermitian Operators)**.** *Hermitian operators possess real eigenvalues.*

Proof. Let's begin with our Hermitian operator A and one of its eigenkets $|u\rangle$ and associated eigenvalue a:

$$A|u\rangle = a|u\rangle$$

If we multiply both sides by $\langle u|$, we get:

$$\langle u|A|u\rangle = \langle u|a|u\rangle = a\langle u|u\rangle$$

Let's now also take the complex conjugate of this:

$$\langle u|A|u\rangle^* = \langle u|A|u\rangle = (a\langle u|u\rangle)^* = a^*\langle u|u\rangle$$

Subtracting these two from each other gives:

$$(a - a^*)\langle u|u\rangle = 0$$

Since we know that $\langle u|u\rangle$ is positive-definite and can only be 0 when $|u\rangle = 0$ (which is not a valid eigenket), this means that $a - a^* = 0$ for this to be true, meaning $a^* = a$. If a number equals its own complex conjugate, its imaginary part must be 0, implying the number is real. Since we took an arbitrary eigenket/eigenvalue pair for an arbitrary Hermitian operator, this means that all Hermitian operators possess *exclusively* real eigenvalues, as desired. ■ □

Next, we have a way to relate eigenbras and eigenkets:

Lemma 2.5 (Equivalence of Left and Right-Eigenvalues)**.** *If $A|u\rangle = a|u\rangle$ yields eigenket $|u\rangle$ and right-eigenvalue a, then $\langle u|A = a\langle u|$ yields eigenbra $\langle u|$ and equal left-eigenvalue a.*

Proof. Starting with our original eigenvalue equation, we can follow the same first step as our previous simplification:

$$A|u\rangle = a|u\rangle \implies \langle u|A|u\rangle = a\langle u|u\rangle$$

Now, dividing out $|u\rangle$ from both sides yields:

$$\langle u|A = a\langle u|$$

as desired. ■ □

Our final simplification is more significant:

Theorem 2.3 (Eigenspaces of Hermitian Operators)**.** *The* eigenspaces *of a Hermitian operator corresponding to distinct eigenvalues are orthogonal.*

Proof. Let's start with two distinct eigenvalue equations for our operator A:

$$A|u\rangle = a|u\rangle, \quad A|u'\rangle = a'|u'\rangle$$

Multiplying the first equation on both sides by $\langle u'|$ and the second by $\langle u|$ gives:

$$\langle u'|A|u\rangle = a\langle u'|u\rangle, \quad \langle u|A|u'\rangle = a'\langle u|u'\rangle$$

If we now subtract the Hermitian conjugate of the second equation from the first, we get:

$$\langle u'|A|u\rangle - \langle u'|A|u\rangle = a\langle u'|u\rangle - a'\langle u'|u\rangle = (a' - a)\langle u'|u\rangle = 0$$

This therefore means that either $a' - a = 0$ or $\langle u'|u\rangle = 0$. If the first is true, then we have $a' = a$, meaning we're dealing with two eigenkets corresponding to the same eigenvalue (in which case we're in one single eigenspace), or else $\langle u'|u\rangle = 0$, which means that eigenkets corresponding to distinct eigenvalues (otherwise belonging to different eigenspaces) are orthogonal, as desired.
■ □

2.3.7 Observables

Will the eigenspaces of an operator always "fill up" the space upon which the operator acts? In other words, can we exclusively describe every ket vector in terms of some eigenkets? Generally, no, but there will be some special cases. An operator whose eigenspaces will "fill up" the space on which the operator acts is called *complete*. This is another instance where we can make tremendous simplifications with Hermitian operators. As it turns out, in finite dimensions, every Hermitian operator is complete. In other words, the sum of the dimensions of the eigenspaces equals the diemsnion of the entire space on which the operator acts.

Now, extending to infinite dimensions, we unfortunately cannot make the same generalization about Hermitian operators. However, we will be much more interested in those Hermitian operators that will be complete in infinite dimensions. We will call such complete Hermitian operators *observables*[7].

2.3.8 Projection Operators

We introduce one final operator which we'll need in the next section, called a *projector*.

[7] The reasoning behind this nomenclature will become apparent in the next chapter, when we start developing physical interpretations for all of this while setting up the postulates underpinning all of quantum theory.

Definition 2.7 (Projection Operator)**.** A *projection operator* (or *projector*) is an observable P that is *idempotent*:

$$P^2 = P$$

It's immediately clear from this definition what eigenvalues projection operators possess:

$$P\,|p\rangle = p\,|p\rangle \implies P^2\,|p\rangle = P\,|p\rangle = p^2\,|p\rangle = p\,|p\rangle \implies p^2 = p \implies \boxed{p = 0, 1}$$

This means that a projection operator will separate a given Hilbert space $\mathcal{E}$ into two orthogonal subspaces, where every ket vector in one space will be annihilated (associated with the eigenvalue 0), while every ket vector in the other will be unaffected (associated with eigenvalue 1). The space associated with the eigenvalue 1 is the space upon which our P *projects*.

Projection operators will be meaningful when it comes to measurements, which are crucial to our study of quantum mechanics, which is why we wanted to mention them here.

2.4 Relevant Mathematical Frameworks

You may have noticed that, throughout the introduction of the relevant formalism, we made references to certain mathematical fields and did not elaborate too much on the framework itself. Physics is fundamentally built upon a mathematical foundation, with different fields/subdisciplines each using their own mathematical frameworks. We address the relevant fundamental frameworks key to quantum mechanics below, with more information on exactly what detail we plan on going into for each, as well as where you can find more information to become better acquainted with the mathematics, should you find that useful for the physics.

2.4.1 Linear Algebra

The entirety of quantum mechanics is fundamentally built on a linear algebra framework, with operators transforming vectors in particular ways to give us meaningful information about a system. As such, a solid foundational understanding of linear algebra is quite helpful if one wants to develop a solid (and intuitive) understanding of quantum mechanics. True proficiency in linear algebra requires a semester-long undergraduate course at the bare minimum, and we do not have the bandwidth to produce an entire linear algebra textbook within a quantum mechanics book, and we do not wish to try. As such, to help with some basic ideas, we provide a very succinct development of some of the key points relevant to our study in a Linear Algebra appendix, and recommend the reader peruse an external textbook (such as the previously-mentioned Sheldon Axler's *Linear Algebra Done Right*) for a more in-depth treatment.

2.4.2 Fourier Analysis

As we will soon discover for ourselves (and as the reader may already be somewhat familiar with), quantum mechanics at its core deals with a wavefunction to describe the behavior of a system. In fact, the Schrodinger equation is itself derived from the wave equation, meaning we will come to care very deeply about periodic functions. A key insight made two centuries ago by Joseph Fourier was that *any* continous function can be decomposed into a series of periodic functions. This is accomplished through the *Fourier transform* that we briefly mentioned at the beginning. A proper treatment of Fourier analysis (and all of its powerful implications) also requires a semester-long undergraduate course at the bare minimum, which we (unsurprisingly) do not have time for, so we won't attempt to develop the full theory either. That being said, there are only a few insights that are immediately relevant to us, which we will detail in additional, shorter appendix.

2.4.3 Vector/Tensor Analysis

As we already saw throughout the development of the formalism, our fundamental unit is a vector on which we will act some operator to yield another vector. Outer products and special chained products of operators (called *tensor products*) will also be of great use to us when we eventually wish to *couple* ket spaces[8]. Tensors are an even more abstract object than the vectors and matrices we've already been dealing with, which will require us to develop new notation to navigate it.

Remark. We've already seen glimpses of this *index notation* in the Kronecker delta!

Since (as you can probably guess) we don't have the time/need to thoroughly develop tensors, we offer a brief development of index notation, how to navigate tensors, and what they are, in yet another appendix. One may also wish to consult (INSERT GOOD TEXTBOOK I DON'T KNOW OF ANY ONES OFF THE TOP OF MY HEAD).

2.4.4 Complex Analysis

The reader may be familiar with some aspects of calculus (generalized formally to a field known as *real analysis*), but what happens when we start dealing with complex-valued functions? Can we perform calculus with them? As it turns out, we can, but the way we do it is not what we may be used to from real analysis. The study of how to deal with such complex functions, known as *complex analysis*, helps us understand how it works. We won't need to worry about this until much later onward, but when we do inevitably require some tools, we will develop them as necessary. We will add another short appendix to help expand on some of

[8] Do not worry about what this means at the moment; we will learn about it much later.

the underlying mathematics, but we strongly encourage the reader to consult a book such as Stein and Shakarchi's *Notes on Complex Analysis* for a deeper understanding.

2.4.5 Final Note

As mentioned at the very beginning of this exceedingly long chapter, there was a lot of math involved. We have made our best attempt to balance outlining the motivation to study certain mathematical concepts without being too technical, but someone who may be unfamiliar with linear algebra may still have struggled to keep up. We understand; we threw a lot at you and promised you that this setup will all have been worth it in the end. To an extent, we hope you will simply take our word for it; that being said, it is completely natural for some of this to not make sense upon first examination. As such, we cannot encourage independent research more. It is the responsibility of any aspiring physicist (or scientist in general) to close any knowledge gaps they feel they may have, and it's unrealistic to expect any singular source to be able to do that for everything there is. Thus, if there is anything you feel we do not cover in sufficient detail (which there has to be by the nature of the content we want to address and the manner in which we wish to present it), we encourage a consultation of the materials listed in our references. You can always come back to these notes after you feel more comfortable with the math; we'll wait. That being said, we hope we can sufficiently motivate the material that the mathematics won't get in the way of a more fundamentally intuitive understanding of what we are developing.

This concludes our development of the fundamental formalism, but not our development of mathematical tools. That said, we will not require nearly as much math development in as dense a space moving forward. Let's finally get into quantum mechanics...

3 The Postulates of Quantum Mechanics

Having established some mathematical formalism, we now have enough tools at our disposal to begin a proper treatment of quantum mechanics. The first step in our journey will be establishing a few fundamental postulates that will make up the foundation of quantum theory. Here, we will motivate the need for such postulates, discuss them in detail, and put them to the test with a real-world experiment conducted at the turn of the previous century to validate our postulates.

3.1 What are Postulates?

Firstly, what even are postulates? Why do we even need them? Formally, a *postulate* (sometimes called an *axiom*) is a statement that we take to be true without proof. In other words, we will treat a postulate as fact without providing rigorous mathematical justification for that fact.

Why do we need these? Can't we just start writing down equations? Well, if we were to try writing down equations, what would they represent? Physics, at its core, aims to explain real-world phenomena through a rigorous mathematical framework. As such, we need a bridge to translate certain physical observations to the formalism we'll use in our theory. From there, once we have a baseline, we can build upwards and derive new reslts to explain new phenomena. That being said, we have to start somewhere, and the easiest place to start is nature. In other words, we can make certain observations of physical phenomena, and use the results of these observations to write down more formal statements that can be further experimentally tested (and ideally confirmed).

Example 3.1 (Euclid's Elements)**.** In mathematics, the need for postulates (and how they arise) is a bit clearer to illustrate, so we will motivate postulates first with *Euclid's Elements*, which form the foundation of the Euclidean geometry taught in high school. They are as follows:

1. A straight line segment can be drawn joining any two points.
2. Any straight line segment can be extended indefinitely in a straight line.
3. Given any straight line segment, a circle can be drawn having the segment as radius and one endpoint as center.
4. All right angles are congruent.
5. If two lines are drawn which intersect a third in such a way that the sum of the inner

angles on one side is less than two right angles, then the two lines inevitably must intersect each other on that side if extended far enough[a].

At face value, these all appear obvious. But how do we *know* that they're true? The short answer is that we just do. We accept that each is a thing that is possible (and it's quite simple to convince ourselves that it's always possible to draw a line through two points, for example), but we make no attempts to actually rigorously *prove* that they're true mathematically. We use these always-true statements to then derive new results that we actually can prove.

[a]This is known as the *parallel postulate*, which turned out to be the most controversial of the postulates, as it only holds in *flat* space. Different variants of curved space led to associated revisions to this parallel postulate, which in turn gave rise to several variants of *non-Euclidean geometry*.

Have we seen any postulates in physics before? As it turns out, yes we have! Although we didn't quite refer to it this way, Newton's first law of motion is, at its core, a statement we make without proof based purely on observation:

> An object in motion will remain in motion (and an object at rest will remain at rest) unless acted upon by an external force.

This is not a law that we prove mathematically; rather, we use it to explicitly define what an *inertial reference frame* is, which we then use to develop more of Newtonian mechanics. We now wish to do the same with quantum mechanics, to connect the formalism we've developed to real-world experimental results.

3.2 The (Incomplete) Postulates

We are now ready to state the postulates of quantum mechanics, in a slightly incomplete form. They are as follows:

Theorem 3.1 (Postulates of Quantum Mechanics (Incomplete))**.** *The incomplete form of the postulates of quantum mechanics are as follows:*

1. *Every physical system is associated with a Hilbert space $\mathcal{E}$, with the vectors of this called* kets*.*
2. *Every* pure state[a] *of a physical system is associated with a definite ray in $\mathcal{E}$.*
3. *Every measurement that can be carried out on the system corresponds to a complete Hermitian operator A, otherwise called an* observable*.*
4. *The possible results of measurement are the eigenvalues A, either discrete $(a_1, a_2, \ldots)$ or continuous $(a(\nu))$.*
5. *In the discrete case, the probability of measuring a particular eigenvalue $A = a_n$ is*

given by:

$$\text{Prob}(A = a_n) = \frac{\langle\psi|P_n|\psi\rangle}{\langle\psi|\psi\rangle};$$

Here, P_n is the projection operator onto eigenspace $\mathcal{E}_n$ corresponding to eigenvalue a_n, and $|\psi\rangle$ is any nonzero ket in the ray representing the state of the system. In the continuous case, the probability of measuring some A as lying somewhere in the interval $I = [a_0, a_1]$ is given by:

$$\text{Prob}(A \in I) = \frac{\langle\psi|P_I|\psi\rangle}{\langle\psi|\psi\rangle};$$

In this case, P_I is the projection operator corresponding to the interval I[b]*.*

6. *After a measurement with some discrete outcome $A = a_n$, the system is represented by the ket $P_n |\psi\rangle$, where $|\psi\rangle$ represents the system before the measurement*[c]*. In the continuous case, after a measurement with some outcome $a_0 \leq A \leq a_1$, the system will be represented by $P_I |\psi\rangle$ after the measurement*[d]*.*

[a] This is a term that we haven't yet defined and won't define until we address the density operator and a more complete form of the postulates.

[b] The numerator here is referred to as the **expectation value** of the operator P_n with respect to $|\psi\rangle$, in many instances shortened to $\langle P_n\rangle$.

[c] In this particular case, the system will be in an eigenstate of A with associated eigenvalue a_n after the measurement.

[d] This is otherwise known as the *collapse postulate*, and is the formal phrasing of the "collapse" of the wavefunction into a new state after a measurement is performed. The physical significance of this wavefunction collapse has been heavily debated for decades, and is still not fully understood.

You may have noticed the term "incomplete" when describing this particular formulation of the postulates. We'll explain exactly what we mean by this (and how we can "complete" these postulates) a bit later.

As we can see, quantum mechanics will rely on measurements of observables to describe the behavior of a physical system. The fifth postulate describes the **probability** of measuring a particular outcome, suggesting that the predictions quantum mechanics will make must be of a **statistical** nature rather than a deterministic one (like Newtonian mechanics). Now, to properly determine these probabilities, we need to repeat the same measurements many times on a large number of identically-prepared systems.

This is a really important point that we will emphasize once more: quantum mechanics as a theory will make **statistical predictions about measurement outcomes on an *ensemble* of many (essentially infinite) identically-prepared systems**. In other words, quantum mechanics tells us that, should we perform the same measurement on a bunch of identical systems many times, we will expect to see a certain outcome a set number of times, with this number given by the expression shown in the 5th postulate. Quantum mechanics does not make any definite claims to how an individual system will behave; rather, it will predict trends. This is one of the major aspects that separates it from a more familiar theory like classical mechanics, where we can make definite, certain predictions on how a given individual system will evolve subject to certain constraints.

Before we make any attempts to "complete" the postulates by properly defining an extra quantity and introducing the 7th and final postulate, let's put the postulates we already wrote down to the test with a real-world experiment to see if they accurately translate between mathematics and reality.

3.2.1 An Application: The Stern-Gerlach Experiment

The postulates we defined are certainly a good starting point, but there are still some unanswered questions that we would really like to understand before continuing. For instance, how can we know precisely what Hilbert space is associated with a given physical system? What mathematical operator would correspond to a definite measurement? Finally, which ray corresponds to a definite (pure) state of a system? While the postulates claim the existence of such Hilbert spaces, operators, and rays, they do not explicitly state how to ascertain their precise values. To do that, we need an actual physical system and an experimental setup where we see for ourselves what these quantities must be. For this, we will consider the **Stern-Gerlach experiment**—conceived by Otto Stern in 1921 and carried out by Walther Gerlach in 1922—which experimentally demonstrated the quantization of the spatial orientation of angular momentum.

Remark. The full technical details behind the theory and execution of the experiment extend beyond the scope of what we aim to cover here. As such, we will omit lengthy discussions of precisely *what* we are measuring and how we're going about doing so. Instead, we urge the reader to focus on how we appropriately utilize the postulates and apply them to a physical system, thereby confirming experimentally what we expect from theory.

Consider the setup illustrated in the figure below. Here, we send out a beam of silver atoms from an oven and measure different components of their *magnetic moment* μ. We first measure the x component of the magnetic moment, denoted μ_x, and pass the beam where the measured value is $\mu_x = +\mu_0$[1] into a second magnet that measures the z component (denoted μ_z).

As a result of this experiment, we will see two beams from the second magnet measurement, with a 50/50 split between $\pm\mu_0$. Let's now use our postulates to analyze this experiment.

Firstly, measuring μ_x appears to have two possible outcomes, namely $\pm\mu_0$. As such, Postulates 3 and 4 suggest that the operator associated with this measurement will have two possible eigenvalues, also $\pm\mu_0$ (we expect μ_y and μ_z to also have two eigenvalues each). Thus, the Hilbert space $\mathcal{E}$ corresponding to this observable must be at least 2-dimensional, which aligns with our experimental measurements of components of the magnetic moment.

[1]Here, $\mu_0 = \frac{e\hbar}{2mc}$ is the *Bohr magneton*; we will not worry about what this is for a while.

To proceed, let us assume that the eigenvalues $\pm\mu_0$ are *nondegenerate* (ie, they each have exactly one associated eigenket). In this case, the eigenspaces are each 1-dimensional, $\mathcal{E}$ is exactly 2-dimensional, and it is spanned by the eigenkets of μ_x with eigenvalues $\pm\mu_0$. We'll similarly expect the space to be spanned by eigenkets of μ_y and μ_z, with the same associated eigenvalues.

Let's denote a normalized eigenvector of μ_x with eigenvalue $+\mu_0$ by the ket $|\mu_x+\rangle$. Let's denote the eigenvector of μ_x with eigenvalue $-\mu_0$ by $|\mu_x-\rangle$, and use a similar convention for the eigenvectors of μ_y and μ_z (namely $\left|\mu_y\pm\right\rangle$ and $|\mu_z\pm\rangle = |\pm\rangle$). We showed in the previous chapter that eigenkets corresponding to distinct eigenvalues are orthogonal, meaning the following is true:

$$\langle\mu_x+|\mu_x-\rangle = \left\langle\mu_y+\middle|\mu_y-\right\rangle = \langle+|-\rangle = 0$$

Now, Postulate 2 claims that the atomic beam's state is described by some state vector at each stage in our experimental apparatus. Additionally, Postulate 6 states that the system will be specified by $|\mu_x+\rangle$ after measuring μ_x with a result of $+\mu_0$, after which it'll be specified by $|\pm\rangle$, depending on the measurement outcome of μ_z. This means that we can represent the state of the beam entering the second magnet (after the μ_x measurement, but before the μ_z measurement) as a linear combination of the μ_z result vectors:

$$|\mu_x+\rangle = c_+ \,|+\rangle + c_- \,|-\rangle$$

Our goal now is to ascertain what these expansion coefficients are, and whether they align with the results of actually performing this experiment. Since the eigenkets $|\pm\rangle$ are orthogonal, we can represent the expansion coefficients via an inner product:

$$c_\pm = \langle\pm|\mu_x+\rangle$$

Postulate 5 then tells us that the probability of measuring $\mu_z = +\mu_0$ is:

$$\text{Prob}(\mu_z = +\mu_0) = \langle\mu_x+|P_+|\mu_x+\rangle$$

Here, $P_+ = |+\rangle\langle+|$ is the projection operator onto the $+\mu_0$ eigenspace of μ_z. Substituting that and setting it equal to our experimentally-determined probability gives: gives:

$$\text{Prob}(\mu_z = +\mu_0) = \langle\mu_x+|+\rangle\,\langle+|\mu_x\rangle = |\,\langle+|\mu_z+\rangle\,|^2 = |c_+|^2 = \frac{1}{2}$$

Taking the square root gives us our expansion coefficient:

$$\therefore c_+ = \frac{1}{\sqrt{2}}e^{i\alpha_1}$$

Here, $e^{i\alpha_1}$ is an unknown *phase factor*. Why do we add it? We add it for the same reason that we add an integration constant $+C$ whenever taking the antiderivative for an indefinite integral: just as how constants disappear under derivatives and must be accounted for in antiderivatives, phase factors disappear when taking the squared norm, meaning they must be reintroduced when taking the square root:

$$|e^{i\alpha}|^2 = e^{i\alpha}e^{-i\alpha} = e^0 = 1$$

Remark. Even though these phase factors cannot be ignored, there are many cases when we can ascertain exactly what they should be, given additional conditions/restrictions on our problem. This is similar to how we can solve for integration constants given additional boundary conditions that may occasionally be imposed on our problem.

Now, since the probability of measuring $\mu_z = -\mu_0$ is also $\frac{1}{2}$, that means we can conclude that c_- must have a similar form, albeit with a different phase factor:

$$\therefore c_- = \frac{1}{\sqrt{2}} e^{i\beta_1}$$

Putting these together gives us our expansion for $|\mu_x +\rangle$:

$$|\mu_x +\rangle = \frac{1}{\sqrt{2}} \left(e^{i\alpha_1} |+\rangle + e^{i\beta_1} |-\rangle \right)$$

This is looking promising, but we sitll have these pesky phase factors to deal with in order for us to have a complete description of the state of the system. There are a few things we can do to ascertain what they are.

Firstly, we can actually get rid of one of these phase factors by "absorbing" it into the other. How? Instead define $|\mu_x +\rangle = |\mu_x +\rangle' e^{i\alpha_1}$ and now multiply our previous equation by $e^{-\alpha_1}$, while also setting $\beta_1' = \beta_1 - \alpha_1$. After dropping the primes (which are merely notational conventions), we are left with:

$$|\mu_x -\rangle = \frac{1}{\sqrt{2}} \left(|+\rangle + e^{i\beta_1} |-\rangle \right)$$

This may look cheesy, and that's because it is. We are abusing the fact that our notation simply represents some arbitrary factor. We used a new definition to shift the phase factor elsewhere, and once we were able to move the entirety of the phase to one ket, we no longer required the primes to distinguish from anything else, so we could safely drop them.

Let's now try to figure out what these phase factors are. Even though we chose distinct variables for these phase factors to avoid assumptions, they indeed are related through the orthogonality condition of these respective eigenkets:

$$\langle \mu_x + | \mu_x - \rangle = \frac{1}{2} \left(1 + e^{i(-\beta_1 + \gamma_1)} \right) = 0 \implies e^{i\beta_1} = -e^{i\gamma_1}$$

This allows us to get rid of γ_1 and write both kets in terms of the same phase factor:

$$|\mu_x \pm\rangle = \frac{1}{\sqrt{2}} \left(|+\rangle \pm e^{i\beta_1} |-\rangle \right)$$

What if we instead opted to measure μ_y at the first stage? As it turns out, the beams emerging from the second magnet will each still respectively carry a 50% probability, meaning we can perform the exact same analysis to conclude:

$$\left|\mu_y\pm\right\rangle = \frac{1}{\sqrt{2}}\left(|+\rangle + e^{i\beta_2}\,|-\rangle\right)$$

We are now ready to write down the operators that represent the components of μ. From the formalism, we have:

$$\mu_x = \mu_0\left(|\mu_x+\rangle\langle\mu_x-| + |\mu_x-\rangle\langle\mu_x+|\right)$$

Using what we discovered previously, we can rewrite this in the $|\pm\rangle$ basis:

$$\mu_x = \mu_0\left(e^{-i\beta_1}\,|+\rangle\langle-| + e^{i\beta_1}\,|-\rangle\langle+|\right)$$

Similarly, we can get the y component:

$$\mu_y = \mu_0\left(e^{-i\beta_2}\,|+\rangle\langle-| + e^{i\beta_2}\,|-\rangle\langle+|\right)$$

Finally, the z component is the most straightforward:

$$\mu_z = \mu_0\left(|+\rangle\langle-| - |-\rangle\langle+|\right)$$

We're at last ready to relate $e^{i\beta_1}$ and $e^{i\beta_2}$ by directly comparing them with a slightly different Stern-Gerlach experimental setup. Instead of measuring μ_z in the second magnet, let's measure μ_y. Experimentally, we still expect to measure $\mu_y = \pm\mu_0$ with equal probability, meaning we have:

$$\frac{1}{2} = |\left\langle\mu_y\pm|\mu_x\pm\right\rangle|^2 = \frac{1}{2}(1\pm\cos(\beta_2-\beta_1)) \implies \beta_2 = \beta_1 \pm \tfrac{\pi}{2} \iff \boxed{e^{i\beta_2} = \pm i e^{i\beta_1}}$$

Thus, we've reduced our unknowns to a single unknown phase (let it be $e^{i\beta_1}$ WLOG[2]). We can finally figure this one out by setting a phase convention for $|-\rangle$. For example, in the $|\pm\rangle$ basis, we can make the matrix elements of either μ_x or μ_y entirely real, but not both. Thus, whichever one we choose to make real will dictate our phase convention.

Remark. Crucially, this demonstrates that it is *impossible* to satisfy the postulates of quantum mechancis using exclusively real numbers, meaning we must employ complex numbers for our results to make sense with the theory we've chosen. This, in a way, means complex numbers are, to an extent, real.

[2] This is an acronym for "without loss of generality". This is used when we are making a specific choice, but emphasize that the choice we make doesn't affect the general outcome, meaning we could've just as easily chosen the other option with the exact same results.

The general convention is to absorb $e^{i\beta_1}$ into the definition of $|-\rangle$, meaning we are left with the following operator representations:

$$\mu_x = \mu_0 \left(|+\rangle\langle-| + |-\rangle\langle+|\right), \quad \mu_y = \mu_0 \left(-i\,|+\rangle\langle-| + i\,|-\rangle\langle+|\right), \quad \mu_z = \mu_0 \left(|+\rangle\langle-| - |-\rangle\langle+|\right)$$

We can write these in a more familiar matrix representation:

$$\mu_x = \mu_0 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mu_y = \mu_0 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \mu_z = \mu_0 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Remark. These are the **Pauli spin matrices**! These are a very important set of matrices and operators that will appear all over the place when we analyze angular momentum and spin, and they also appear as fundamental single-qubit gates in quantum information science. We will not explain their significance here, but rest assured, this is not the last time we will see them.

3.2.2 Commuting Observables

Our discussion thus far has been correct and does indeed yield expected theoretical results, but it assumed one crucial thing: that the eigenspaces of μ_x, μ_y, and μ_z are nondegenerate. How do we know this is the case? What would happen if it isn't? Since we have no Hilbert space or Hermitian matrix to fall back on (given that we're building them from the postulates themselves), we must rely on other tools to address them. This is a more advanced topic that, while important, isn't really worth discussing in much detail here. The purpose of this section was to show how we can derive crucial quantum-mechanical results exclusively from the postulates and relevant experimental results. Since we feel we have adequately shown this, we leave the remainder of the nuances up to the reader to learn more about. We encourage them to consult Notes 2 of UC Berkeley prof. Robert Littlejohn, which can be found in the References.

3.2.3 The Density Operator, Briefly

The majority of our previous analysis was predicated on the assumption that we are accurately able to describe the quantum state of each beam in our experimental apparatus by a definite state vector (in other words, what's known as a *pure state*). Is this necessarily the case? Well, the quantum state of the beam that first emerges from the first magnetic measurement is known and can properly be described; what about the state of the beam that first enters the measurement apparatus from the oven?

A long answer (explored more deeply in Littlejohn Notes 3) made incredibly short is **no**. In general, the state of a particular system may not be specified by a definite state vector, meaning we need more complicated tools to accurately describe the statistics at play. Let's very briefly construct the solution to this problem. Imagine instead a family of normalized kets that depend on some continuous parameter(s) λ, denoted $|\psi(\lambda)\rangle$, and let's assume that this parameter is distributed according to some probability distribution $f(\lambda)$ such that:

$$f(\lambda) \geq 0 \quad \text{and} \quad \int d\lambda f(\lambda) = 1$$

Then, the expectation value of any arbitrary operator A can be rewritten in terms of this new distribution:

$$\langle A \rangle = \int d\lambda f(\lambda) \langle \psi(\lambda) | A | \psi(\lambda) \rangle$$

The discrete case is a lot simpler; here, we enumerate the states $|\psi_i\rangle$ and our probability distribution is now discretized:

$$\sum_i f_i = 1$$

Similarly, our expectation value shifts from an integral to a discrete sum over the probabilities f_i:

$$\langle A \rangle = \sum_i f_i \langle \psi_i | A | \psi_i \rangle$$

Let's now introduce the *density operator* to help us systematize our analysis:

Definition 3.1 (Density Operator)**.** The *density operator*, denoted ρ, is defined like so in terms of the given kets $|\psi\rangle$ and probabily distributions f:

$$\rho = \int d\lambda f(\lambda) |\psi(\lambda)\rangle\langle\psi(\lambda)| \text{ (continuous)}, \quad \rho = \sum_i f_i |\psi_i\rangle\langle\psi_i| \text{ (discrete)}$$

We can now combine the discrete and continuous versions of our expectation value into one unique expression in terms of ρ:

$$\langle A \rangle = \mathrm{tr}(\rho A)$$

Here, tr represents the *trace* of the associated matrix.

Now, when the state of a system can be completely represented by some definite state vector (as is the case for the beam that enters the second magnetic measurement in our Stern-Gerlach apparatus), then the density operator is a simple outer product in terms of that exact state:

$$\rho = |\psi\rangle\langle\psi|$$

In this case, we say that the state of our system is a *pure state*, and our postulates apply as written. However, if we have any statistical uncertainty in the state vector that would represent our system (and thus can no longer say that this singular state vector would have an associated probability of 100%), we say our system is in a *mixed state*. These are going to be much more prevalent in real life, and require more nuance in our postulate formulation.

Remark. There is some subtlety in one what mean by "definite state vector" when discussing pure vs. mixed states. For example, if our system is represented by two kets that only differ by some phase factor and we choose to attach a 50% measurement probability to each of them, that's in fact a pure state, since these vectors are indeed related. This subtlety will not be addressed any further than this here, because we will often not havve to deal with mixed states and the density operator at this level of study.

The density operator has a variety of associated properties that we will not worry about here (but may feature in appendices and that are definitely addressed in some of our References), so we will forego any additional discussion of it. We will simply use it to revise our postulates for them to be slightly more complete and precise. We present the revisions below.

Theorem 3.2 (Postulate of Quantum Mechanics (Complete Form))**.** *The slightly revised postulates of quantum mechanics are as follows:*

1. *Every physical system is associated with a Hilbert space $\mathcal{E}$.*
2. *Every state of a physical system is associated with a* density operator *ρ acting on $\mathcal{E}$, which is a Hermitian, positive-semidefinite operator of unit trace ($\operatorname{tr}\rho = 1$).*
3. *Every measurement process that can be carried out on the system corresponds to a complete Hermitian operator A.*
4. *The possible results of the measurement are the eigenvalues of A, either the discrete eigenvalues a_i or the continuous ones $a(\nu)$.*
5. *The average value measured for the operator A is $\operatorname{tr}(\rho A)$. The probability of measuring a given eigenvalue for both the discrete and continuous case is given by:*

$$\operatorname{Prob}(A = a_n) = \operatorname{tr}(\rho P_n) \text{ (discrete)}, \quad \operatorname{Prob}(A \in I) = \operatorname{tr}(\rho P_I)$$

   In the discrete case, P_n is the projection operator onto the eigenspace $\mathcal{E}_n$ with associated eigenvalue a_n. In the continuous case, P_I is the projection operator corresponding to the interval $I = [a_0, a_1]$.
6. *If a system is described by a density operator ρ (which we assume to be normalized). Then, after a measurement of the operator A producing some (possibly degenerate) eigenvalue a_n, the system will be described by the (normalized) density operator*

$$\rho' = \frac{P_n \rho P_n}{\operatorname{tr}(\rho P_n)},$$

   where P_n is the projector onto the n-th eigenspace of A.

Now, we have a more complete set of postulates that we can apply to any kind of physical system. Whenever we deal with a system that can be represented by a pure state (which will be often in the beginning), we can default to the earlier, incomplete variant and not worry about the density operator too much. Nevertheless, we felt it was important to explain what it was and (intuitively) why we need it.

Remark. This may have been fairly confusing, and we completely understand. Sometimes, we will state important corrections to certain principles for the sake of completeness and total transparency, but at this stage, the simplified variants will be more than sufficient to develop the theory. For more detailed discussions of these principles, await a graduate course in quantum mechanics and consult J. J. Sakurai's *Modern Quantum Mechanics*.

3.3 Quantum Mechanics from Classical

We laid the theoretical framework underpinning quantum mechanics in the formalism and applied to the real world with the postulates. At long last, we are able to start developing some physics.

3.3.1 Why all the Setup?

Many undergraduate courses in quantum mechanics will skip over all of this formal development at the beginning in favor of jumping straight into working with the wavefunction and utilizing the Schrodinger equation to solve basic problems. This choice is typically made to avoid adding needless complexity and to prevent the technical mathematical details from getting in the way of a basic physical intuition. As a matter of fact, this is precisely how we learned quantum mechanics initially, before supplementing the finer mathematical formalism in later graduate courses.

We considered following suit and also omitting these details to jump straight into the physics, but decided to include this for a few reasons. Primarily, we did this because **this is not a university course on quantum mechanics**. We are not bound by any time constraints and do not need to adhere to any exlicit schedule, so the reader can take as much time as is required for everything to make sense. Since we don't need to move through a unit within a week or two, there's no longer a pressing need to omit certain details in the name of expediting the material.

Remark. This is not in any way suggesting that the way undergraduate courses structure quantum mechanics is wrong. Rather, since we are not bound by the same restrictions that an undergraduate course is, we can take certain liberties in material presentation.

Since we don't have the same time restrictions, we can build up the material a bit more rigorously and mathematically. Even though this rigorous development requires a certain level of mathematical maturity and is ultimately more challenging for a first reader than jumping straight into the physics would be, we feel that it is extremely satisfying to see all of the operators and equations we use pop out directly from first principles and experimental results. This way, we don't need to simply state the form of an operator without justification simply because we want to practice using it. Thus, we hope you have beared with us while we laid all the pieces in place.

As we warned at the beginning of the previous chapter, there is a *lot* of mathematics involved in the development of a formal framework for a physical theory. If the reader wishes to gain a more solid mathematical footing before attempting to understand certain parts of the way we present the material, that's totally fine! We will wait here for you to return. It is our hope, though, that we do a sufficiently good job of presenting whatever mathematics is necessary throughout the relevant chapters and accompanying appendices at the back.

3.3.2 The Position Operator

Let's start with figuring out how to describe a particle's position in quantum mechanics. For simplicity, we will start with the 1-dimensional case, then address 3 dimensions later on. Let's also assume that a particle's position in space (call it x) can be measured, and these measurement results are continuous (since space isn't discretized). Let's then denote the operator associated with measuring the position $\hat{x}$ and any eigenkets of the position $|x\rangle$. Thus, our eigenvalue problem takes the following form:

$$\hat{x}\,|x\rangle = x\,|x\rangle$$

We will see much more of this hat notation moving forward; an operator will typically be denoted with a hat on top of it to indicate that it is an operator rather than a variable. The main exceptions will be those operators labeled with *capital* letters, since those are harder to confuse for variables or other mathematical objects.

Since we are dealing with a continuous spectrum and nondegenerate eigenkets (positional values can only correspond to a unique position), the resolution of the identity for this operator takes the following form:

$$\int_{-\infty}^{\infty} dx\,|x\rangle\langle x| = 1$$

Now, let $|\psi\rangle$ represent a normalized pure state for our system, and $I = [x_0, x_1]$ be some interval on the x-axis. In this case, the projector onto the interval looks like this:

$$P_I = \int_{x_0}^{x_1} dx\,|x\rangle\langle x|$$

Then, the probability postulate tells us that the probability of measuring some positional value within this interval is given by the following expectation value:

$$\text{Prob}(x_0 \le x \le x_1) = \langle\psi|P_I|\psi\rangle = \int_{x_0}^{x_1} dx\,\langle\psi|x\rangle\,\langle x|\psi\rangle$$

Now, that last expression looks like we're taking some squared norm. To make our lives easier (and to make this expression look more like an integral expression wherein we're actually integrating a function), let's finally define the *positional wavefunction* $\psi(x)$:

Definition 3.2 (Positional Wavefunction)**.** The *wavefunction*, denoted $\psi(x)$, encodes information about the position of a particle and is given by the following inner product:

$$\psi(x) = \langle x|\psi\rangle$$

Then, our probability becomes:

$$\text{Prob}(x_0 \leq x \leq x_1) = \int_{x_0}^{x_1} dx|\psi(x)|^2 = \int_{x_0}^{x_1} \psi^*(x)\psi(x)dx$$

Here, we see that the squared norm of the wavefunction encodes the probability density of finding the particle at a given point in space. Now, we finally have a way to go from kets to functions!

Can we go the other way? No problem! If we multiply both sides of the resolution of the identity by $|\psi\rangle$, we get a representation of the ket:

$$|\psi\rangle = \int dx\,|x\rangle\,\langle x|\psi\rangle \implies \boxed{|\psi\rangle = \int dx\,|x\rangle\,\psi(x)}$$

Now that we have a way to convert between ket formalism and functional notation, we also have a way to describe the inner product between two kets $|\psi\rangle$ and $|\phi\rangle$ by once again inserting the resolution of the identity between them:

$$\langle\psi|\phi\rangle = \int dx\,\langle\psi|x\rangle\,\langle x|\phi\rangle = \int \psi^*(x)\phi(x)dx$$

This formalism extends quite easily to three dimensions, where we simply have to consider three separate positional components (x, y, z). Since we will mainly deal with 1-dimensional problems in the next chapter, we'll save the discussion of the 3D generalization for later.

3.3.3 The Momentum Operator

Lastly, let's take a look at the momentum. Although position is important, objects tend to move, so it's also helpful to quantify how we should go about measuring this change in motion, namely the momentum.

Remark. I did not, in fact, take a look at the momentum. Oopsies.

3.3.4 Final Note

As mentioned earlier, much of what appears in this chapter is generally omitted in a first course on quantum mechanics in favor of jumping straight into the physics and trying to describe systems with new equations. Perhaps this may have been easier, but at the end of the day, it is entirely about the journey rather than the destination. Since we are not beholden to any schedule and are attempting to organize these notes in a way that we felt works best, we have chosen to take the road less traveled. There is a light at the end of this tunnel, however; we are at last done with the difficult setup. The following chapters will take a much lighter approach that may be more akin to an undergraduate perspective, albeit with some formalism sprinkled in every now and then.

Let's use our newly-acquired tools to finally start describing some basic one-dimensional systems!

4 1D Wave Mechanics

Now that we have a solid foundation of the formalism of quantum mechanics, we can begin our exploration of where these formalisms lead us. In this section, we will introduce the wavefunction, see how it is used in the time independent Schrodinger equation to extract the allowed states in our system, and explore some classical problems studied in most quantum mechanics books. Finally, we will look at how to incorporate time dependence and transform a time-independent solution $\psi(x)$ into a time-dependent $\Psi(x,t)$.

4.1 The Wavefunction

As mentioned in [REF formalisms], the fundamental object in our study of quantum mechanics is the **wavefunction**, denoted $\psi(x,t)$ or $\Psi(x,t)$, which is a vector in our Hilbert space $\mathscr{H}$. This is also sometimes denoted as the **state** of our system, and is represented in bra-ket notation by $|\psi\rangle$. To begin our discussion, we will focus on 1D wave functions; that is, wavefunctions of the form $\psi(x,t)$ (so only having one position coordinate), and then later generalize this notion to multiple dimensions. This is summarized in the box below:

The state of a quantum system is denoted by its wavefunction ψ, which lives in a Hilbert space $\mathscr{H}$, and is represented in functional form by $\psi(x)$ or in its vector form by $|\psi\rangle$.

Recall that ψ itself isn't something that can be directly observed, but it contains all the information about the quantum system. What we do observe is $|\psi|^2$, which represents the probability density of measuring a particular observable (assuming of course, that ψ is expressed in the basis of the corresponding observable). As $|\psi|^2$ is a probability density, it must follow the law of total probability, which is written as:

$$\int_{-\infty}^{\infty} |\psi(x,t)|^2 \; dx = \int_{-\infty}^{\infty} \psi(x)\psi^*(x) \; dx = 1$$

at all t, with the ψ^* being the complex conjugate of ψ. Intuitively, this should also make sense: this is essentially saying that the probability you measure *some* value of your observable is guaranteed.

Note that this does NOT imply that ψ is necessarily integrable, and that's perfectly fine! Remember that what we physically observe is the probability $|\psi|^2$ and not ψ, so there is no requirement on what ψ needs to be. Formally, ψ belongs to a class of functions which are considered *square-integrable*, namely the property that $|\psi|^2$ must be integrable.

4.2 Time Independent Schrodinger Equation

Now, we will take a look at how we actually write down the wavefunction ψ. To begin, recall that from [REF formalisms] that ψ can be expressed in many different ways, depending on the basis we choose. One particular basis of interest to us is the *energy* basis, which are eigenstates of the Hamiltonian operator $\hat{H}$. These are of particular interest to us since at a high level, one primary property we care about in our quantum system is the energy of our particles. Now, since ψ is written as a linear combination of these eigenstates, the next logical step is to solve the eigenvalue equation for H to find them. This equation is known as the **time-independent Schrodinger equation (TISE)**, which we introduce below:

Theorem 4.1 (Time-Independent Schrodinger Equation)**.** *Given a Hamiltonian $\hat{H}$, its energy eigenstates are given by the eigenvalue equation:*

$$\hat{H}\left|\psi\right\rangle = E\left|\psi\right\rangle$$

This equation is at the heart of quantum mechanics – everything we do from here on out is all in the name of solving this equation. As it turns out, this equation is surprisingly complex, and we shall see that in this chapter.

In light of this, there are still some things we can do. To begin, we should unpack the Hamiltonian operator $\hat{H}$, and write it down in a "nicer" form. First, we go back to the classical (non-quantum) definition of the Hamiltonian as the sum of kinetic and potential energies of a system:

$$H = T + V$$

The kinetic energy, written in terms of the momentum p, is:

$$T = \frac{p^2}{2m}$$

So the Hamiltonian can be written as:

$$H = \frac{p^2}{2m} + V(x)$$

For now (and for most of this book), we will consider time-independent potentials $V(x)$. To transform this Hamiltonian into the quantum language, we promote all observables to operators (motivated by our postulate), so this means we promote p to $\hat{p}$, and x to $\hat{x}$.
The position operator is $\hat{X} = x$, and the momentum operator is $\hat{p} = -i\hbar\frac{\partial}{\partial x}$.[1] So, the time-independent Schrodinger equation now reads:

$$\left(-\frac{i\hbar}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\left|\psi\right\rangle = E\left|\psi\right\rangle$$

[1] See [INSERT NOTE HERE] for a more in-depth explanation as to why the momentum and position operator are expressed in this way. For now, we will accept this as fact.

and now the path forward is clear: for different systems, we substitute in the appropriate potential $V(x)$, and solve the differential equation for ψ, and out comes the energy eigenstates!

4.3 Fundamental Problems

Armed with the time-independent Schrodinger equation, let's go ahead and see some simple but illustrative examples of $V(x)$ for which ψ is analytically solvable. We first begin with the simplest case for V, and work our way to more complicated potentials.

4.3.1 The Free Particle

What's the simplest form of $V(x)$? Well, $V(x) = 0$ of course! This case is known as the *free particle*, which we will study below. In this case, the time-independent Schrodinger equation reads:

$$-\frac{i\hbar}{2m}\frac{\partial^2}{\partial x^2}\psi = E\psi$$

this is a simple second order ODE, whose solutions are complex exponentials. To solve this, let k be defined as:

$$k = \frac{\sqrt{2mE}}{\hbar}$$

so the differential equation now reads:

$$\frac{\partial^2}{\partial x^2}\psi = -k^2\psi$$

which has a well-known solution of the form:

$$\psi(x) = Ae^{ikx} + Be^{-ikx}$$

which makes sense – these are just plane waves travelling through space unimpeded, which you'd expect if you don't have a potential $V(x)$. In particular, this solution is a superposition of a wave travelling to the right with positive k and one to the left with negative k.
However, this solution has a problem: it violates normalizability! If we try to check the normalization:

$$\int_{-\infty}^{\infty} \left(Ae^{ikx} + Be^{-ikx}\right)\left(A^*e^{-ikx} + B^*e^{ikx}\right) = \int_{-\infty}^{\infty} |A|^2 + |B|^2 + AB^*e^{2ikx} + A^*Be^{-2ikx}\, dx = \infty$$

which diverges to infinity. So what does this mean? Well, the only natural conclusion is that such a wavefunction is inadmissible, and therefore a free particle with a definite energy E

cannot exist. Instead, the solution is to consider a linear combination of a *range* of energies, represented as an integral:

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ikx} \, dk \tag{4.1}$$

where $\phi(k)$ essentially represents the "weight", or contribution that the wave with momentum k conributes to the overall $\psi(x)$. Those that are familiar with Fourier analysis will recognize this equation immediately, and point out that $\psi(x)$ and $\phi(k)$ are related by the Fourier transform:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x) e^{-ikx} \, dx$$

And because of this fact, this means that $\psi(x)$ and $\phi(k)$ are actully equivalent descriptions of the same state! This doesn't resolve our initial problem though: how do we find $\phi(k)$? The trick lies in the fact that $\phi(k)$ has no explicit time-dependence, so we can pick a time which we know the form of $\psi(x)$ (which must be normalized), and solve for $\phi(k)$ there. Generally, this time is chosen to be at $t = 0$, so:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x, 0) e^{-ikx} \, dx$$

which you can then plug into Equation 4.1 to get $\psi(x)$. In essence, because the Fourier transform doesn't touch the normalization condition at all (it just provides a means to transform from position to frequency space), once you start with a normalized wavefunction, it will always be normalized.

4.3.2 The Infinite Square Well

With $V(x) = 0$ solved, let's move forward and look at our first non-trivial $V(x)$, defined by the piecewise function:

$$V(x) = \begin{cases} 0 & 0 \le x \le a \\ \infty & \text{otherwise} \end{cases}$$

this is called the *infinite-square-well*, appropriately named since the graphical representation of the potential is a well whose walls are infinitely tall:

```
\begin{tikzpicture}[scale=1.5]
    \draw [semithick, stealth-stealth] (-1, 3) node[left] {$\infty$} -- (-1, 0) node[lef
    -- (1, 0) node[right] {$a$} -- (1, 3) node[right] {$V(x)$};
\end{tikzpicture}
```

To solve this differential equation, we will consider $V(x)$ in the interval $[0, a]$ separately from everywhere else, then tie the solutions together. Luckily for us, both solutions are relatively simple. between $[0, a]$, since $V(x) = 0$, then we revert back to the above case of the free particle, where our solution is just travelling plane waves. For reasons that will be apparent later, we will choose the trigonometric representation of a travelling wave:

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

This is an equivalent way to representing a travelling wave, and the conversion is given to us by Euler's formula:

$$e^{ikx} = \cos(kx) + i\sin(kx)$$

Now, outside of $[0, a]$, the potential is infinite, so the only sensible solution for ψ is $\psi = 0$.

Okay, so we have the solutions in these separate regimes, how do we tie them together? The key is to realize that $\psi(x)$ must be *continuous*, since x (the position) is a continuous variable and $|\psi(x)|^2$ must represent the probability density in position. So, since the wavefunction is identically zero outside the well, it must approach zero from inside the well toward the boundaries:

$$\lim_{x\to 0}\psi(x) = \lim_{x\to a}\psi(x) = 0$$

First, we handle the $x = 0$ case since that's easy:

$$\psi(0) = A\sin(0) + B\cos(0) \implies B = 0$$

Now for the $x = a$ case:

$$0 = \psi(a) = A\sin(ka)$$

so then ka must be an integer multiple of π, hence $ka = n\pi$ or generally written as $k_n = \frac{n\pi}{a}$, the subscript n-th solution. And here we see a very intriguing property fall out of the mathematics: the simple restriction of our wavefunction in space *forces* our wavefunction to belong to a discrete family of functions, parametrized by the integer n. As a consequence, this also quantizes the allowable energies:

$$E_n = \frac{p^2}{2m} = \frac{\hbar^2 k_n^2}{2m}$$

This is the first result that fundamentally differs from classical notions of thinking, as energy is always thought to be a continuum in classical physics. Finally, all we have to do now is to find the normalization constant A, which we do by enforcing the law of total probability:

$$\int_{-\infty}^{\infty} |A|^2 \sin^2 kx\ dx = |A|^2\frac{a}{2} \implies |A| = \sqrt{\frac{2}{a}}$$

Therefore, the full solution to the infinite square well is the set of wavefunctions ψ_n:

$$\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)$$

Hopefully through this example you can start to see how the idea of quantization falls out very naturally from the simple restriction that ψ must be continuous: it's an example of how quantum mechanics is fundamentally different from classical mechanics, and yet it makes extremely accurate descriptions of the phenomena around us.

4.3.3 Delta Function Potentials

In this section, we will consider $V(x)$ of a very particular form:

$$V(x) = -a\delta(x)$$

Here, $\delta(x)$ refers to the Dirac delta function, which is quite a funny object. It's not a "function" in the traditional sense, and instead defined as a function whose value is zero everywhere but integrates to 1 over the entire real line. In this sense, it's possible to represent the delta function as:

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & \text{elsewhere} \end{cases}$$

while also satisfying:

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1$$

there is no true function that simultaneously satisfies these properties, hence the delta function isn't really considered a function.[2] However, it often makes an appearance in theoretical and experimental physics since many phenomena around us can be modeled approximately as delta functions.

Just like all of the potentials we've seen so far, let's throw this $V(x)$ into the differential equation:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} - a\delta(x)\psi = E\psi$$

Now, let's consider two separate cases: when $x < 0$ and $x > 0$. In both these cases, $\delta(x) = 0$, so therefore our differential equation reduces to:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = E\psi \implies \frac{\partial^2\psi}{\partial x^2} = -\frac{2mE}{\hbar^2}\psi$$

Now it's time to mention a detail we haven't really talked about until now: and that is the sign of E that we're dealing with. As you'll notice from the differential equation above, the sign of E determines whether the sign of the prefactor, which impacts the kind of solutions our differential equation yields. In particular, when $E > 0$, then we admit *scattering* solutions, and when $E < 0$ we admit *decaying* solutions.[3]

[2] Formally, $\delta(x)$ is called a *distribution* rather than a function, but we won't worry too much about this distinction. It's good enough for our purposes to treat $\delta(x)$ as a function that is zero everywhere except at $x = 0$ where it equals infinity, and that its integral over all space is equal to one.

[3] In case you're wondering why we haven't dealt with the sign of E until now, it's because we didn't need to. In the free particle example, a particle with negative energy and admits a decaying solution would blow up at either ∞ or $-\infty$, which are not normalizable and thus inadmissible. A negative energy state isn't admissible in the infinite square well potential either, since there's no way for an exponentially decaying function to be identically zero as is required to match the boundary condition on either side of the well.

The decaying solutions are slightly easier to deal with, so let's tackle those first. In this case, $E < 0$ so the prefactor on ψ is positive, hence we can write:

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{2mE}{\hbar^2} = \kappa^2 \psi$$

where:

$$\kappa = \frac{\sqrt{-2mE}}{\hbar} \tag{4.2}$$

remember, κ is still positive here since E is negative. Just like the free particle solution, both the positive and negative exponential are valid solutions to the differential equation:

$$\psi = Ae^{\kappa x} + Be^{-\kappa x}$$

and here's where our physical boundary conditions come into play. First, let's consider the $x < 0$ case. Here, we require that ψ doesn't blow up as $x \to -\infty$, which is only possible if the negative exponential vanishes. Likewise, when $x > 0$ we don't want ψ to blow up as $x \to \infty$, hence the positive exponential term vanishes. Put put together, we have the following "pieces" of our wavefunction:

$$\psi_L = Ae^{\kappa x} \quad \psi_R = Be^{-\kappa x}$$

Now, we need to join the two together to get one complete wavefunction. To do this, we enforce our notion of continuity, which is to say that ψ_L and ψ_R must attain the same value at $x = 0$. This is easy; since they both have the same exponential, this is just telling us that $A = B$. So, the wavefunction looks like:

$$\psi = \begin{cases} Ae^{\kappa x} & x < 0 \\ Ae^{-\kappa x} & x > 0 \end{cases}$$

To find A, we enforce normalization. To do this, notice that ψ is symmetric about $x = 0$, so we can write the integral as:

$$1 = \int_{-\infty}^{\infty} |\psi|^2 \, dx = 2\int_0^{\infty} |A|^2 e^{-2\kappa x} \, dx = \frac{|A|^2}{\kappa}$$

So the solution is $|A|^2 = \kappa$, or $A = \sqrt{\kappa}$, and our wavefunction is then:

$$\psi = \sqrt{\kappa} e^{-\kappa|x|}$$

Note that I combined the piecewise expression into a single one with the use of the absolute value on x. But what would be the energy of such a wavefunction? To find it, we have to look more closely at the behavior of ψ around $x = 0$. To find the energy, we need to impose an additional condition that we haven't encountered yet: continuity of ψ'.

First off, we should try and understand the reason why continuity in ψ' is needed, of which there are two. The first comes directly from our time-independent Schrodinger equation. Recall that the equation reads:

$$\left(-\frac{i\hbar}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)|\psi\rangle = E\,|\psi\rangle$$

isolating the term on the left, we have:

$$-\frac{i\hbar}{2m}\frac{\partial^2\psi}{\partial x^2} = (E - V(x))\psi$$

since ψ is continuous then the right side is continuous, which means the left must be also. If ψ'' is continuous, then this implies that ψ' must be differentiable, hence it must be continuous.

But what if $V(x)$ is not continuous? We've seen two examples already where $V(x)$ is not continuous, so what then? As it turns out, even when $V(x)$ is discontinuous, as long as $V(x)$ is not *infinite*, then continuity in $d\psi/dx$ is still guaranteed, the proof of which we outline below:

Theorem 4.2. *For a given potential $V(x)$, as long as $V(x)$ is finite, the wavefunction ψ and its derivative ψ' is always continuous.*

Proof. Consider an arbitrary $V(x)$. Then, Schrodinger's equation reads:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(x) = E\psi(x)$$

we can now integrate this equation around an arbitrary point $[a-\epsilon, a+\epsilon]$:

$$-\frac{\hbar^2}{2m}\int_{a-\epsilon}^{a+\epsilon}\frac{\partial^2\psi}{\partial x^2}\,dx + \int_{a-\epsilon}^{a+\epsilon}V(x)\psi(x)\,dx = E\int_{a-\epsilon}^{a+\epsilon}\psi(x)\,dx$$

The first term is just the derivative $\psi'(a+\epsilon) - \psi'(a-\epsilon)$, and the term on the right hand side vanishes as $\epsilon \to 0$. So that leaves us with the equation:

$$\psi'(a+\epsilon) - \psi'(a-\epsilon) = \frac{2m}{\hbar^2}\int_{a-\epsilon}^{a+\epsilon}V(x)\psi(x)\,dx$$

and now we come to the heart of the argument: if $V(x)$ is finite, then the integral on the right vanishes, and we get:

$$\psi'(a+\epsilon) - \psi'(a-\epsilon) = 0$$

as $\epsilon \to 0$, which is the definition of continuity in ψ'. However, if $V(x)$ is infinite, then this equality fails. Hence, we've proven that ψ' is always continuous, as long as $V(x)$ is not infinite. $\square$

This is also why we didn't have to consider continuity in ψ' when dealing with the infinite square well. Now we return to the delta function. In this case, since $V(x)$ is not finite, a discontinuity in ψ' is allowed, but the discontinuity at $x = 0$ will tell us more about the allowed energy:

$$\psi'(\epsilon) - \psi'(-\epsilon) = -\frac{2m}{\hbar^2}\int_{-\epsilon}^{\epsilon}a\delta(x)\psi(x) = -\frac{2ma}{\hbar^2}\psi(0) = -\frac{2ma}{\hbar^2}A \tag{4.3}$$

The derivative of ψ is:

$$\psi' = \begin{cases} A\kappa e^{\kappa x} & x < 0 \\ -A\kappa e^{-\kappa x} & x > 0 \end{cases}$$

and as $x \to 0$, then $\psi' \to A\kappa$ for $x < 0$ and $A\kappa$ for $x > 0$. Substituting these values into Equation 4.3 gives us:

$$-2A\kappa = -\frac{2ma}{\hbar^2}A \implies \kappa = \frac{ma}{\hbar^2} \tag{4.4}$$

which means we can finally solve for E! Recall all the way back (Equation 4.2) that we defined E in terms of κ. But now since we know κ in terms of other constants, we can finally back out E:

$$E = -\frac{ma^2}{2\hbar^2}$$

and so finally, we have the full solution for the delta function potential, for $E < 0$:

$$\psi(x) = \frac{\sqrt{ma}}{\hbar}e^{-ma|x|/\hbar^2}, \quad E = -\frac{ma^2}{2\hbar^2}$$

Now, we have to deal with the scattering states $E > 0$. In this case, the solutions at $x < 0$ and $x > 0$ both admit plane wave solutions, one traveling to the left and another to the right:

$$\psi(x) = Ae^{-ikx} + Be^{ikx}, \quad k = \frac{2mE}{\hbar}$$

The coefficients at this moment are undetermined, so we have four constants to deal with:

$$\psi_L(x) = Ae^{-ikx} + Be^{ikx} \quad \psi_R(x) = Ce^{-ikx} + De^{ikx}$$

We now know how to combine them into one big equation: we first enforce continuity at $x = 0$, which gives us the equation $A + B = C + D$. Now, we enforce continuity in the derivatives. They are:

$$\begin{aligned} \psi_L'(0) &= ik(B - A) \\ \psi_R'(0) &= ik(D - C) \end{aligned}$$

Employing the same trick as Equation 4.3, we get:

$$\psi'(\epsilon) - \psi'(-\epsilon) = ik(D - C) - ik(B - A) = \frac{2ma}{\hbar^2}\psi(0) = \frac{2ma}{\hbar^2}(A + B)$$

We choose to use $\psi_L(0) = A + B$ for the last equality; the logic works the same if you use ψ_R. There is a more compact way to write this equality, if you let $\beta = \frac{ma}{\hbar^2 k}$ then you have:

$$C - D = A(2i\beta + 1) + B(2i\beta - 1)$$

So we have the following two equations for A, B, C and D:

$$\begin{aligned} A + B &= C + D \\ D - C &= B(1 - 2i\beta) - A(1 + 2i\beta) \end{aligned}$$

But this is a big problem! We have 4 constants but only two equations, so we'll never be able to fully solve for all constants. There is a solution though – and the key is to appeal to our intuition about what system we're trying to describe. Say we're sending in a wave from the left. Then, it would not make sense to consider a left-traveling wave in ψ_R, so that means that $C = 0$, and from here we solve for A and D in terms of B. If you work out the algebra, you get the following:

$$A = -B\frac{i\beta}{1 + i\beta}, \quad D = \frac{B}{1 + i\beta}$$

So what do we make of these solutions? By solving for the coefficients A and D, they tell you the amplitude of the reflected and transmitted wave when you send in a wave with amplitude B from the left. In particular, we can now calculate quantities such as the reflection coefficient, which gives you the probability that a wave is reflected:

$$R = \frac{|A|^2}{|B|^2} = \frac{\left|-B\frac{i\beta}{1+i\beta}\right|^2}{|B|^2} = \frac{\beta^2}{1 + \beta^2}$$

and similarly, we can calculate a transmission coefficient:

$$T = \frac{|D|^2}{|B|^2} = \frac{\left|\frac{B}{1+i\beta}\right|^2}{|B|^2} = \frac{1}{1 + \beta^2}$$

what's nice is that we can verify that the reflection and transmission add up to 1:

$$R + T = \frac{\beta^2}{1 + \beta^2} + \frac{1}{1 + \beta^2} = 1$$

which makes sense – the wave is either transmitted or reflected, without loss. If you now bring back the substitution for β that we did earlier, we can write these in terms of the energy itself:

$$R = \frac{1}{1 + (2\hbar^2 E/ma^2)}, \quad T = \frac{1}{1 + (ma^2/2\hbar^2 E)}$$

and that's the full solution to the delta barrier! Firstly, you can already see that this derivation is *very* non-classical: it predicts that a potential well has the chance to scatter a particle

At this point, let's talk briefly about the delta potential *barrier*, which is basically the same potential but positive: $V(x) = a\delta(x)$. While it may seem like we need to do the math all over again, we actually don't since switching the sign only changes the behavior at $x = 0$, so our scattering state remains the same. What does change though is the bound state. To see why, recall that for the bound state, κ was defined as (Equation 4.2):

$$\kappa = \frac{\sqrt{-2mE}}{\hbar}$$

which was guaranteed to be *real* and *positive*, since $E < 0$. However, if we look at our other definition of κ in terms of a and m (Equation 4.4):

$$\kappa = \frac{ma}{\hbar^2}$$

and so if $a < 0$, this solution for κ becomes inconsistent with the first, and hence such a bound state cannot exist. However, the scattering solutions are fine because we don't have an equivalent restriction on k.

Remark. It is interesting to note that because the transmission and reflection are dependent on a^2, these quantities remain the same regardless of a well or a barrier! This is quite striking – for one, just the mere fact that two different potentials give you the same reflection and transmission is fascinating. Moreover, the fact that a potential well can induce a reflection is also fascinating. Classically, this would be the equivalent of throwing a ball over a well and it having a chance to bounce right back at you, but in quantum mechanics, such a phenomenon is totally allowed!

4.3.4 The Potential Step

Here, we will consider a potential represented by a wall of finite height:

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & x > 0 \end{cases}$$

There are two different kind of states to consider, $0 < E < V_0$ and $E > V_0$, and the $E < 0$ state does not exist. Let's consider the $E < V_0$ state first. To the left of the barrier, we have a scattering state, but to the right we have a *decaying* state, since $E - V_0$ is negative. This is because inside the barrier, Schrodinger's equation reads:

$$\frac{\partial^2 \psi}{\partial x^2} = (E - V_0)\psi$$

so following the same substitution for k as in Section 4.3.1, you get:

$$k = \frac{\sqrt{2m(E - V_0)}}{\hbar} = \frac{i\sqrt{2m(V_0 - E)}}{\hbar}$$

since k is purely imaginary, this turns the oscillatory e^{ikx} into a decaying exponential. Combining this with the oscillatory solution to the left of the barrier gives you the following solution for ψ:

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < 0 \\ Ce^{\kappa x} + De^{-\kappa x} & x > 0 \end{cases}$$

with k and κ being defined as:

$$k = \frac{\sqrt{2mE}}{\hbar} \quad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar^2}$$

Again, we're faced with a situation where continuity alone doesn't give us enough equations to fully solve, and have to be more specific about the problem to eliminate some variables. Firstly, let's say that our wave is coming in from the left. This would mean that inside the barrier, an exponentially *growing* solution would not make sense, and therefore we can set $C = 0$. Now, we have enough equations to solve for B and D in terms of A. In particular, at $x = 0$, the continuity constraints in ψ and ψ' gives us the following system of equations:

$$A + B = D$$
$$ik(A - B) = -\kappa D$$

Now, expressing B and D in terms of A (remember, A is our incident wave amplitude, so it makes sense to solve in terms of A) just requires some algebra, and ultimately gets you the relations:

$$B = \frac{ik + \kappa}{ik - \kappa} A, \quad D = \frac{2ik}{ik - \kappa} A$$

Here, we can calculate the reflection coefficient:

$$R = \frac{|B|^2}{|A|^2} = \left| \frac{ik + \kappa}{ik - \kappa} \right|^2 = 1$$

so this tells us that the entire amplitude of the wave is reflected, which makes sense since the barrier is infinitely thick, so any wave will be fully reflected back. As we'll see in Section 4.3.5, when the barrier is only finitely thick you will get a transmitted wave, called *tunnelling*.

You may be wondering: if the entire wave is reflected, then why is the wavefunction nonzero inside the barrier? Doesn't this imply that there is a nonzero probability for the particle to *be* inside the barrier, and thus not be reflected?

The answer to this question is quite subtle. Firstly, decaying solutions don't carry any energy, as their probability current turns out to be zero. Hence, all the energy must be reflected back. But, perhaps this isn't a satisfying answer, in which case there another solution. Suppose you *were* able to measure the particle inside the barrier. Well, the decaying solution has a characteristic length of $\frac{1}{\kappa}$, so your variance in position is $\Delta x \sim \frac{1}{\kappa}$. But, by the uncertainty relation, you find that:

$$\Delta p_x \geq \frac{\hbar}{\Delta x} = \hbar\kappa = \sqrt{2m(V_0 - E)}$$

and plugging this into the energy:

$$\Delta E = \frac{(\Delta p_x)^2}{2m} \geq V_0 - E$$

and this tells us that by measuring your particle inside the barrier, you've inadvertently *scattered* your particle, and it's no longer in the barrier! So, it's physically impossible to measure your particle to be inside the barrier, no matter how careful you are.

Now we consider the second solution, the $E > V_0$ case. In this case, both $x < 0$ and $x > 0$ admit scattering solutions, but with different wavenumbers:

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < 0 \\ Ce^{ikx} + De^{-ikx} & x > 0 \end{cases}, \qquad k = \frac{\sqrt{2mE}}{\hbar}, \quad k' = \frac{\sqrt{2m(E - V_0)}}{\hbar}$$

again, we impose a physical constraint, of considering a particle incident from the left. This kills the left travelling wave in $x > 0$, so $D = 0$. Using the continuity constraints again, we get the following relation:

$$B = \frac{kk'}{k + k'}A \quad C = \frac{2k}{k + k'}A$$

This algebra follows very similarly from the previous examples, so we'll skip showing it here. With this, we can now calculate the reflection and transmission:

$$R = \frac{|B|^2}{|A|^2} = \frac{\left[1 - \left(1 - \frac{V_0}{E}\right)^{1/2}\right]^2}{\left[1 + \left(1 - \frac{V_0}{E}\right)^{1/2}\right]^2} \quad T = \frac{4\left(1 - \frac{V_0}{E}\right)^{1/2}}{\left[1 + \left(1 - \frac{V_0}{E}\right)^{1/2}\right]^2}$$

and you should verify on your own that $R + T = 1$. Just like the delta potential well, we find that even though the energy of your particle can be higher than the step, you *still* get a reflection regardless, which is a fundamentally non-classical result.

4.3.5 The Potential Barrier

This potential is similar to that of the potential step, but this time it's a finitely long wall:

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & 0 < x < a \\ 0 & x > a \end{cases}$$

visually, it looks like this:

```
\begin{tikzpicture}[scale=2]
    \draw (-1, 0) -- (0, 0) node[below] {$0$} -- (0, 1) node[left] {$V_0$} -- (1, 1) -- (1, 
\end{tikzpicture}
```

Here, there are two cases we need to consider: $E > V_0$ and $0 < E < V_0$. For $E < 0$, we get an infinitely decaying solution, which is to say that $\psi = 0$ in that case. Let's start with the $0 < E < V_0$ case, which as we mentioned before, will lead to a tunnelling situation. We'll split up the wavefunctions into three regions:

1. To the left of the wall: $x < 0$
2. Above the wall: $0 < x < a$
3. To the right of the wall: $x > a$

So the respective wavefunctions we solve for will be labeled ψ_1, ψ_2 and ψ_3 according to the above numbering. Starting off with $x < 0$, we have the same plane wave solution as before:

$$\psi_1(x) = Ae^{ikx} + Be^{-ikx}$$

with $k = \frac{\sqrt{2mE}}{\hbar}$ as before. The same goes for the right:

$$\psi_3(x) = Ce^{ikx} + De^{-ikx}$$

Now let's look at the Schrodinger equation for $0 < x < a$:

$$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi + V_0\psi = E\psi$$

Moving the V_0 to the right gives:

$$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi = (E - V_0)\psi$$

And since $E < V_0$ here, then we have exponentially decaying solutions inside the wall:

$$\psi_2(x) = Fe^{\kappa x} + Ge^{-\kappa x}$$

with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. Unlike the potential step though, we can't simply eliminate the exponentially growing solution by setting $F = 0$. Because the barrier isn't infinitely long, the exponentially growing solution is allowed. However, we can still eliminate D, by considering a particle incident from the left.

Now, because we have split ψ into three wavefunctions, we need to stitch them together by enforcing continuity in ψ and ψ' at $x = 0$ *and* $x = a$. The $x = 0$ conditions are:

$$A + B = F + G$$
$$ik(A - B) = \kappa(F - G)$$

and similarly, the $x = a$ ones are:

$$\begin{aligned} Ce^{ika} &= Fe^{\kappa a} + Ge^{-\kappa a} \\ ike^{ika} &= \kappa(Fe^{\kappa a} - Ge^{-\kappa a}) \end{aligned}$$

From here, we can then solve for $\frac{B}{A}$ and $\frac{C}{A}$, which will give us the reflection and transmission coefficients. At this point, the algebra also gets quite long so we won't go into too much detail, but there is a relatively "nice" solution involving hyperbolic sine and cosine:

$$R = \left[1 + \frac{4E(V_0 - E)}{V_0^2 \sinh(ka)}\right], \quad T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)}\right]$$

Now, we turn our attention to the $E > V_0$ case. Here, all three regions have scattering solutions, but k is different above the barrier because of V_0:

$$\psi = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < 0 \\ Fe^{ik'x} + Ge^{-ik'x} & 0 < x < a \\ Ce^{ikx} + De^{-ikx} & x > a \end{cases}$$

Again, follow the same algebra and you'll get:

$$R = \left[1 + \frac{4E(E - V_0)}{V_0^2 \sin^2(k'a)}\right]^{-1} \qquad T = \left[1 + \frac{V_0^2 \sin^2(k'a)}{4E(E - V_0)}\right]^{-1}$$

Here, we want you to focus on the physics and not the algebra so we won't go into detail, but it really does work in the same way we've always been doing it: consider a left-incident wave, set $D = 0$, then eliminate F and G and solve for $\frac{B}{A}$ and $\frac{C}{A}$. Finally, square these and you get your reflection and transmission coefficient.

Speaking of focusing on the physics, notice that because the transmission term has a $\sin^2(k'a)$ term, it means that there are some configurations where the $T = 1$ and $R = 0$, hence you get perfect transmission!

4.3.6 Finite Square Well

Finally, the last potential we will look at in this chapter is the finite square well. This is similar to the infinite square well, but this time our walls are finitely tall. As we'll see, this gives us a first glimpse into why the Schrodinger equation is difficult to solve analytically. Mathematically, we will express this potential as:

$$V(x) = \begin{cases} -V_0 & -a < x < a \\ 0 & \text{otherwise} \end{cases}$$

and here's a visual picture of what the well looks like:

```
\begin{tikzpicture}[scale=1.5]
    \draw (-2, 2) node[left] {$0$} -- (-1, 2) -- (-1, 0) node[below] {$-a$} -- (1, 0) no
\end{tikzpicture}
```

There are two cases we have to consider: the $-V_0 < E < 0$ bound state, and $E > 0$, the scattering state.
Starting with the first case, let's again break up ψ into two parts: inside and outside the well. Inside the well, the Schrodinger equation is written as:

$$\frac{\partial^2 \psi}{\partial x^2} + \alpha^2 \psi = 0$$

with α defined as:

$$\alpha = \left[-\frac{2m}{\hbar^2}(V_0 + E)\right]^{1/2}$$

This is a scattering solution, since the energy of the particle is larger than $-V_0$, so it will give us oscillatory solutions. For reasons that will become clear in a moment, we will write ψ in this region in terms of sines and cosines:

$$\psi = A\cos(\alpha x) + B\sin(\alpha x), |x| < a$$

Outside the well, we have decaying solutions since $E < 0$. The Schrodinger equation in this region reads as:

$$\frac{\partial^2\psi}{\partial x^2} - \beta\psi(x) = 0$$

and β is:

$$\beta = \left(-\frac{2mE}{\hbar^2}\right)^{1/2}$$

Now, we will leverage the following theorem to make our lives easier:

Theorem 4.3 (Wavefunction for Symmetric Potentials)**.** *If the potential $V(x)$ satisfies $V(x) = V(-x)$, then its wavefunction also satisfies $\psi(x) = \psi(-x)$.*

Proof. Suppose first that $\psi(x)$ satisfies the Schrodinger equation for $V(x)$. That is, $\psi(x)$ is a solution to:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2} + V(x)\psi(x) = E\psi(x) \tag{4.5}$$

Now, consider $\psi(-x)$:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(-x)}{\partial x^2} + V(-x)\psi(-x) = E\psi(-x)$$

since $V(-x) = V(x)$, then:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(-x)}{\partial x^2} + V(x)\psi(-x) = E\psi(-x)$$

and since this differential equation is the same as the original equation (Equation 4.5), then $\psi(-x)$ is also a solution. Since this is the case, then any even and odd linear combination of these will also be a solution. That is, the functions:

$$\phi(x) = \psi(x) + \psi(-x)$$
$$\phi'(x) = \psi(x) - \psi(-x)$$

are also solutions. Therefore, we can always take ψ to be even or odd. □

In our case, the finite square well indeed does satisfy this property, and so we can focus on looking for even and odd solutions. This makes our lives much easier, since it means that we only need to consider one set of boundary conditions at $x = a$, instead of two.

Let's start by considering ψ even. Then, the wavefunction is:

$$\psi(x) = \begin{cases} B\cos(\alpha x) & 0 < x < a \\ Ce^{-\beta x} & |x| > a \end{cases}$$

now, imposing boundary conditions at $x = a$ gives:

$$\psi : \;\; A\cos(\alpha x) = Ce^{-\beta a}$$
$$\psi' : \;\; -\alpha A\sin(\alpha a) = -\beta Ce^{-\beta a}$$

dividing one equation by the other gives:

$$\alpha\tan(\alpha a) = \beta$$

similarly, for the odd solutions, we kill the $B\cos(\alpha x)$ term in $0 < x < a$ and the decaying solution remains the same. Applying the same boundary conditions gives you the equation:

$$\alpha\cot(\alpha a) = -\beta$$

both of these equations represent the condition for the allowed energies of the system. That is, for a given a and V_0, these equations tell you which energies E are allowed. Once you figure that out, you can then normalize by enforcing that it integrates to 1. What's interesting about these equations is that they're not analytically solvable, so you can only solve this numerically.

Remark. Hopefully, by working out the these bound states you can get a better appreciation for why the Schrodinger equation is typically a very difficult equation to solve analytically. The potential we're dealing with here is hardly anything complicated – it's just a simple square well of finite depth, yet we're already having to resort to numerical methods to get the allowed energies.

As you can guess, this becomes a general trend as we move toward more complicated potentials. Rarely are any of these potentials analytically solvable, so we have to resort to numerical methods or find a way to approximate the solutions. This latter idea is the central idea of the second part of this book, [INSERT PART 2 REFERENCE].

Now let's deal with the scattering solution. This ends up working out similarly to the delta potential barrier and well, in that the math actually works out the same way. For the left incident particle, we have:

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < -a \\ Fe^{ik'x} + Ge^{-ik'x} & -a < x < a \\ Ce^{ikx} & x > a \end{cases}$$

there is no left travelling wave at $x > a$ since we are considering a left-incident wave. Notice that this is exactly the same situation as the finite potential barrier, except k' is *slightly* different, in that we make the swap $V_0 \to -V_0$. All the algebra remains the same, so we can just copy over the reflection and transmission coefficient we got from there and make this simple edit to V_0:[4]

$$R = \left[1 + \frac{4E(E+V_0)}{V_0^2 \sin^2(2k'a)}\right]^{-1} \qquad T = \left[1 + \frac{V_0^2 \sin^2(2k'a)}{4E(V_0+E)}\right]^{-1}$$

4.4 Time-Dependent Schrodinger Equation

So far, we've been dealing with the time *independent* Schrodinger equation, which as its name suggests, does not include time as a parameter. In this section, we will look at how we transform a solution to the time-independent solution into one that satisfies time dependence. While this may seem complicated at first, the trick actually turns out to be very simple.

4.4.1 Separable solutions

To begin, let's take a look at the time-dependent Schrodinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

Because we deal with time-independent potentials, it means that we can actually solve this equation by method of *separation of variables*. That is, we look for solutions of the form:

$$\Psi(x,t) = \psi(x)\phi(t)$$

solving for $\psi(x)$ was pretty much all that we've been doing in this chapter so far. Now, given a $\psi(x)$, what is the appropriate $\phi(t)$ to tack on so that $\Psi(x,t)$ is a solution to the time-dependent equation? Let's first plug this separable solution back into the Schrodinger equation:

$$i\hbar\psi \frac{\mathrm{d}\phi}{\mathrm{d}t} = -\frac{\hbar^2}{2m} \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\phi + V\psi\phi$$

dividing through by $\psi\phi$:

$$i\hbar \frac{1}{\phi} \frac{\mathrm{d}\phi}{\mathrm{d}t} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + V \tag{4.6}$$

now comes a crucial argument: the left hand side is a function of t, while the right side is a function of x. Therefore, in order for this equation to hold both sides **must** be constant.

[4]We also have to make the change $a \to 2a$ compared to the square barrier since a was the width of the barrier, but a as defined in the finite well is only half the width of the well.

Otherwise, you could change one side of this equation without changing the other, violating the equality. Therefore, we can set both sides to be constant:

$$i\hbar \frac{1}{\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t} = E \implies \frac{\mathrm{d}\phi}{\mathrm{d}t} = -\frac{iE}{\hbar}\phi$$

this is now a simple differential equation for ϕ, which gives solutions:

$$\phi(t) = e^{-iEt/\hbar}$$

and that's our time dependence! All you have to do now is throw this onto every solution to the spatial $\psi(x)$, and you've got your general solution $\Psi(x,t)$. As a side note, setting the right hand side of Equation 4.6 to E gives you the time-independent Schrodinger equation!

At this point, you may also be wondering if we've covered *all* possible solutions this way. After all, the condition that $\Psi(x,t)$ be separable is a fairly restrictive one. As it turns out, whenever the Hamiltonian is time-independent and $\hat{H}$ admits a complete set of energy eigenfunctions, then the separable solutions are enough to describe any general solution. Here, completeness refers to the condition that any square-integrable function (i.e. any function in our Hilbert space $\mathcal{H}$) can be written as a linear combination of these eigenfunctions. Since every solution $\Psi(x,t)$ must be square-integrable, then this completeness condition is all we need to guarantee that we've covered all solutions.

Remark. This completeness condition is actually not as restrictive as you might think – all the Hamiltonians we will deal with in this book satisfy this completeness, even though we'll be looking at some pretty complicated potentials later on!

References